

Organizational Conditions for AI-Driven Firm Performance

A Systematic Review of Empirical Evidence, 2015–2026

Bachelor's Thesis

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Abstract

Artificial intelligence (AI) is available to every firm, and a technology every rival can buy is a weak basis for advantage. Whether and when AI investments pay off is an empirical question. This thesis asks under what conditions firm-level AI investments translate into competitive advantage and superior firm performance. It answers the question with a systematic literature review: a documented Scopus search over the publication window 2015 to 2026 identified 432 records, which a journal-quality filter and two screening stages reduced to 67 empirical studies, each coded by two independent coders against a fixed scheme. Across the 67 studies, the evidence leans positive, with 39 reporting positive average effects, and it is conditional almost throughout: 63 studies identify at least one condition that decides whether the investment reaches its outcome. The review orders these conditions into six groups: complementary investments, organizational capabilities, leadership and governance, firm characteristics, environmental conditions, and the dose, credibility, and timing of the investment itself. Competitive advantage, kept analytically separate from performance throughout, is measured in only 15 studies and almost never as persistence against imitation. The findings support theories that locate AI's payoff in what surrounds the technology rather than in the technology itself, and they mark persistence as the field's largest open question.

Keywords: artificial intelligence; AI investment; firm performance; competitive advantage; systematic literature review; organizational conditions

Contents

1	Introduction	3
2	Theoretical Background	5
2.1	Competitive advantage and firm performance	5
2.2	Artificial intelligence as a general-purpose technology	6
2.3	The resource-based view and the imitability problem	7
2.4	Automation and augmentation	8
2.5	Situated AI	9
3	Method	11
3.1	Review design	11
3.2	Search strategy	11
3.3	Selection criteria	12
3.4	Screening process	13
3.5	Coding	14
4	Results	16
4.1	Descriptive mapping of the evidence base	16
4.2	Synthesis of conditions	19
4.2.1	Complementary investments and resources	20
4.2.2	Organizational capabilities and human capital as the channel	22
4.2.3	Leadership, governance, and management	25
4.2.4	Firm characteristics and market position	26
4.2.5	Environmental conditions	28
4.2.6	Dose, credibility, and timing	30
5	Discussion	33
5.1	The conditional answer	33
5.2	Reading the evidence through the four lenses	34
5.3	Implications for practice	36
5.4	Limitations	37
5.5	Future research	38

6 Conclusion	39
References	40
Appendix	47

Chapter 1

Introduction

Whether investing in artificial intelligence (AI) makes a firm more competitive is, in Kemp’s words, a theoretical puzzle (Kemp, 2024, p. 618)¹. The promise is large: intelligent algorithms are expected to automate a substantial share of work and to open new products and product-development processes. The same literature lists strategic obstacles: AI can be myopic and hard for managers to steer, and, built on explicit knowledge, it resembles a general-purpose technology (Kemp, 2024, p. 618)². A technology every competitor can buy is a poor candidate for an advantage. What firms get out of buying it is an empirical question.

The empirical evidence has grown quickly and does not agree with itself. On German panel data, AI use raises productivity for the yes/no measure and the intensity measure alike (Czarnitzki et al., 2023, pp. 189, 201)³. Growth from AI investments concentrates among the larger firms, and industry concentration rises alongside (Babina et al., 2024, p. 1)⁴. Investors, on average, react to AI investment announcements with a drop of 1.77 percent on the announcement day, and firms with weak IT bases or poor credit ratings lose more (Lui et al., 2022, p. 373)⁵. And where adoption is observed over time, it destroys old

¹[Kemp (2024), p. 618] “The prospect of using artificial intelligence (AI) to establish competitive advantages presents a theoretical puzzle.”

²[Kemp (2024), p. 618] “Despite this promise, a growing body of research has highlighted that AI may present substantial strategic obstacles. AI may be myopic (Balasubramanian, Ye, & Xu, 2022), incapable of perceiving interdependencies within a firm (Raisch & Krakowski, 2021), and recalcitrant to managerial control (Murray, Rhymer, & Sirmon, 2021).” / “In addition, AI is a form of explicit knowledge (Broussard, 2018; Shrestha, He, Puranam, & von Krogh, 2021), and resembles a general-purpose technology (Teece, 2018).”

³[Czarnitzki et al. (2023), p. 189] “We analyze cross-section as well as panel data from the German part of the European Commission’s Community Innovation Survey (CIS).” / [p. 201] “We found that employing AI technologies has a positive and significant impact on firm productivity. In particular, we showed that both the use of AI and the intensity with which firms exploit the potential of AI significantly increase both sales and value-added.”

⁴[Babina et al. (2024), p. 1] “AI-powered growth concentrates among larger firms and is associated with higher industry concentration.”

⁵[Lui et al. (2022), p. 373] “The stock prices of the firms decrease by 1.77% on the day of the announcement. Nonmanufacturing firms and firms with weak information technology capabilities or low credit ratings suffer a more negative impact compared with other firms.”

capabilities while creating new ones at the human-machine boundary (Krakowski et al., 2023, p. 1425)⁶. Positive, negative, and conditional findings stand side by side, often for the same outcome. The disagreement itself points at the question worth asking: under which circumstances do the investments pay off.

That is the question this thesis asks: *under what conditions do firm-level AI investments translate into competitive advantage and superior firm performance?* The two outcome constructs are kept analytically separate throughout, because a firm can improve its performance without holding anything a rival cannot copy; the theoretical grounds for that separation are laid out in Chapter 2.

The thesis answers the question with a systematic literature review of the empirical evidence, searched over the publication window 2015 to 2026. A documented Scopus query identified 432 records, which a journal-quality filter, title-abstract screening, and full-text screening reduced to a final corpus of 67 empirical studies; every study was coded along a fixed scheme by two independent coders, and the resulting coding table is the review’s core deliverable. Chapter 2 condenses the four theoretical lenses used to interpret the evidence: general-purpose technologies, the resource-based view, automation and augmentation, and situated AI. Chapter 3 documents the search, screening, and coding protocol. Chapter 4 presents the findings in two parts, a descriptive mapping of the evidence base and a synthesis of the conditions the studies identify, ordered into six groups. Chapter 5 interprets the findings against the theoretical lenses and names the limits of what the reviewed literature can show, and Chapter 6 concludes.

⁶[Krakowski et al. (2023), p. 1425] “AI adoption triggers interrelated substitution and complementation dynamics, which make humans’ traditional competitive capabilities obsolete, while creating new sources of persistent heterogeneity when humans interact with chess engines.”

Chapter 2

Theoretical Background

2.1 Competitive advantage and firm performance

Competitive advantage is measured in 15 of the 67 studies reviewed here, and the remaining 52 report performance outcomes only. Barney separates the two constructs by what rivals are able to do: a firm holds a competitive advantage while it runs a value-creating strategy no rival, actual or potential, is running at the same time, and that advantage counts as sustained only once those rivals also cannot reproduce what the strategy yields (Barney, 1991, p. 102)¹. Sustained in that sense says nothing about elapsed time. It asks whether the advantage is still there after rivals have given up trying to copy it (Barney, 1991, p. 102)². This review therefore keeps the two apart throughout, because a firm can raise its margins without holding anything a rival is unable to copy.

Where both constructs appear, the separation is visible in the survey items. Chatterjee et al. ask about rivals directly: one of their competitive-advantage items has respondents affirm that successful implementation of AI-based customer relationship management “will help an organization to win over its competitor which has not yet implemented AI-CRM for B2B relationship management” (Chatterjee et al., 2021, p. 215)³. Their performance items stay inside the firm and ask whether the same system improves “operational efficiency” and speeds up decisions (Chatterjee et al., 2021, p. 215)⁴.

¹[Barney (1991), p. 102] “In this article, a firm is said to have a *competitive advantage* when it is implementing a value creating strategy not simultaneously being implemented by any current or potential competitors. A firm is said to have a *sustained competitive advantage* when it is implementing a value creating strategy not simultaneously being implemented by any current or potential competitors *and* when these other firms are unable to duplicate the benefits of this strategy.”

²[Barney (1991), p. 102] “Rather, whether or not a competitive advantage is sustained depends upon the possibility of competitive duplication. Following Lippman and Rumelt (1982) and Rumelt (1984), a competitive advantage is sustained only if it continues to exist after efforts to duplicate that advantage have ceased.”

³[Chatterjee et al. (2021), p. 215] “I think that successful implementation of AI-CRM will help an organization to win over its competitor which has not yet implemented AI-CRM for B2B relationship management.”

⁴[Chatterjee et al. (2021), p. 215] “I believe that successful implementation of AI-CRM will help the organization to improve its operational efficiency.” / “I think AI-CRM helps in quick decision making

Mehta et al. give competitive advantage a position between adoption and results: they test it as the channel through which AI adoption reaches performance, and predict a positive effect on performance in its own right (Mehta et al., 2025, p. 550)⁵.

Not every study that names competitive advantage builds its own instrument. Hossain et al. measure sustained competitive advantage with items carried over from earlier work (Hossain et al., 2022, p. 247)⁶. The construct label alone says little about what a study measured; the items decide.

2.2 Artificial intelligence as a general-purpose technology

Economists file AI under general-purpose technologies (GPTs), and that classification already predicts an uneven payoff. A GPT earns the label because it makes “new and complementary production methods” available, which can raise output over time (Czarnitzki et al., 2023, p. 188)⁷. Those complements are the expensive part. Before output moves, a firm has to redesign how work is organised, retrain staff, rework its software and accumulate managerial experience, and because none of that accumulation is booked as an asset, measured productivity dips before it climbs (Brynjolfsson et al., 2021, pp. 333–334)⁸. The lag is years, not quarters.

Two studies in this corpus show how differently that plays out at firm level. On German innovation-survey panel data, Czarnitzki et al. measure a clear productivity gain from AI use, for the yes/no measure and for the intensity measure alike (Czarnitzki et al., 2023, pp. 189, 201)⁹. Fontanelli et al. ask a different question of French firms and find

which helps the organization to improve its operational performance.”

⁵[Mehta et al. (2025), p. 550] “H8. Competitive advantage has a significant and positive impact upon firm performance.” / “H9. Competitive advantage has a mediating influence on adoption of AI technologies and firm performance.”

⁶[Hossain et al. (2022), p. 247] “sustained competitive advantage items were taken from Cao et al. (2019).”

⁷[Czarnitzki et al. (2023), p. 188] “The main feature of general-purpose technology is to enable new and complementary production methods that may increase productivity over time (Bresnahan and Trajtenberg 1995; Bresnahan et al., 2002; Brynjolfsson and Hitt 2003; Cardona et al., 2013).”

⁸[Brynjolfsson et al. (2021), pp. 333–334] “realizing their potential also requires large intangible investments and a fundamental rethinking of the organization of production itself. Firms must create new business processes, develop managerial experience, train workers, patch software, and build other intangibles. This raises productivity measurement issues because intangible investments are not readily tallied on a balance sheet or in the national accounts.” / [p. 334] “The error in measured total factor productivity growth therefore follows a J-curve shape, initially dipping while the investment rate in unmeasured capital is larger than the investment rate in other types of capital, then rising as growing intangible stocks begin to contribute to measured production.”

⁹[Czarnitzki et al. (2023), p. 189] “We analyze cross-section as well as panel data from the German part of the European Commission’s Community Innovation Survey (CIS).” / [p. 201] “We found that employing AI technologies has a positive and significant impact on firm productivity. In particular, we showed that both the use of AI and the intensity with which firms exploit the potential of AI significantly increase both sales and value-added.”

that AI use raises the *volatility* of productivity growth; the effect sits with firms that buy AI from outside providers, and it weakens where more of the payroll is made up of ICT engineers and technicians (Fontanelli et al., 2025, pp. 1–2)¹⁰. Across the two results, the reading this review takes forward is that the same AI investment can stand behind either outcome; what differs between the firms is the complements.

2.3 The resource-based view and the imitability problem

Across the studies reviewed here, the resource-based view (RBV) is the recurring reason given for expecting AI to pay off. Barney sets four conditions on a resource before it can hold an advantage in place: it has to be valuable, rare among competitors and imperfectly imitable, and no substitute may do the same strategic job (Barney, 1991, pp. 105–106)¹¹. Alwakid and Dahri restate the four conditions as the VRIN acronym (Alwakid & Dahri, 2025, p. 14)¹². That is a shorthand Barney himself comes close to when he lists “rareness, inimitability, non-substitutability” among the characteristics that can qualify a firm attribute as a source of advantage, adding that the attribute becomes a resource only once it exploits an opportunity or neutralizes a threat (Barney, 1991, p. 106)¹³. Hossain et al. put the same argument in terms of what a firm does with what it owns, and tie the long-run case to holdings rivals cannot match (Hossain et al., 2022, pp. 241–242)¹⁴.

Purchased AI fails that test, and the RBV literature says so plainly. Krakowski et al. note that AI costs almost nothing to reproduce and that little stands in the way of copying it, so where AI substitutes a human capability, the RBV predicts that the

¹⁰[Fontanelli et al. (2025), p. 1] “This paper shifts the focus and investigates the impact of predictive AI on the volatility of firms’ productivity growth rates.” / “Using firm-level data from the 2019 French ICT survey, we provide robust evidence that AI use is associated with increased volatility.” / [pp. 1–2] “Our results show that heightened volatility is concentrated among AI buyers, whereas firms that develop AI internally experience no such association. Finally, we find that the AI-volatility link among ‘AI buyers’ is mitigated in firms with a higher share of ICT engineers and technicians, suggesting that AI’s successful integration requires complementary human capital.”

¹¹[Barney (1991), pp. 105–106] “To have this potential, a firm resource must have four attributes: (a) it must be valuable, in the sense that it exploit opportunities and/or neutralizes threats in a firm’s environment, (b) it must be rare among a firm’s current and potential competition, (c) it must be imperfectly imitable, and (d) there cannot be strategically equivalent substitutes for this resource that are valuable but neither rare or imperfectly imitable.” Typographical errors as in the original.

¹²[Alwakid & Dahri (2025), p. 14] “Traditionally, RBV posits that firms achieve sustained competitive advantage by strategically deploying valuable, rare, inimitable, and non-substitutable (VRIN) resources (Barney, 1991; Wang & Zhang, 2025).”

¹³[Barney (1991), p. 106] “Firm attributes may have the other characteristics that could qualify them as sources of competitive advantage (e.g., rareness, inimitability, non-substitutability), but these attributes only become resources when they exploit opportunities or neutralize threats in a firm’s environment.”

¹⁴[Hossain et al. (2022), pp. 241–242] “The resource-based view (RBV) portrays how a firm can optimally use its resources, which are non-substitutable, unique and valuable for achieving sustained competitive advantage (Wernerfelt, 1984).”

advantage resting on that capability will erode (Krakowski et al., 2023, p. 1426)¹⁵. Their own evidence relocates the advantage rather than denying it: what resists imitation is the combination a firm builds, not either side of it (Krakowski et al., 2023, p. 1446)¹⁶.

Two consequences follow for this review. The RBV speaks to competitive advantage and not to performance, so a study reporting higher margins has not thereby shown that anything is defensible against rivals; the separation of the two constructs in Section 2.1 is a theoretical requirement, not only a coding convention. And if the resource itself is imitable, the variation worth studying lies in what surrounds it, which is why the conditions under which AI pays off are treated here as the object of analysis rather than as controls.

2.4 Automation and augmentation

The same investment can be deployed in two ways, and the strategy literature keeps them apart. Raisch and Krakowski define automation as machines taking a task over from a human, and augmentation as humans and machines working on a task together (Raisch & Krakowski, 2021, p. 192)¹⁷. Their argument against the popular advice to prefer augmentation is that the two cannot be held apart in practice. For any single task at any single moment the firm picks one approach, and that choice makes the two contradictory; widen the frame to several tasks over several years and each turns out to produce the other, which makes them interdependent (Raisch & Krakowski, 2021, pp. 195–197)¹⁸.

Corpus studies pick the distinction up as a design choice with different returns. Zebec and Indihar Štemberger treat the two as the use cases through which AI adoption reaches performance, automation paying off in efficiency and augmentation in what people and machines manage jointly (Zebec & Indihar Štemberger, 2024, p. 7)¹⁹. Their model then

¹⁵[Krakowski et al. (2023), p. 1426] “When AI substitutes humans’ cognitive capabilities, the RBV expects the advantage that these capabilities provided to erode (Peteraf & Bergen, 2003). This is because, as a technology resource, AI has close to zero marginal reproduction costs and few imitation barriers (Brynjolfsson & McAfee, 2014, p. 31).”

¹⁶[Krakowski et al. (2023), p. 1446] “Our findings show that humans remain sources of sustainable advantage by combining and integrating their capabilities with machine capabilities into uniquely complementary bundles (Argyres & Zenger, 2012).” / “the locus of sustainable advantage is neither human nor machine capabilities, but what humans do with these capabilities.”

¹⁷[Raisch & Krakowski (2021), p. 192] “Whereas automation implies that machines take over a human task, augmentation means that humans collaborate closely with machines to perform a task.”

¹⁸[Raisch & Krakowski (2021), p. 195] “Automation and augmentation are contradictory, because organizations choose either one or the other approach to address a given task at a specific point in time.” / “If we increase our analysis’s temporal scale (from one point in time to its evolution over time) and spatial scale (from one to multiple tasks), we comprehend that, in the management domain, the two AI applications are not only contradictory but also interdependent.” / [p. 196] “Consequently, augmentation may enable a transition to automation over time.” / [p. 197] “Automation in one task ‘spills over,’ enabling adjacent tasks’ augmentation.”

¹⁹[Zebec & Indihar Štemberger (2024), p. 7] “Automation involves machines taking over tasks from humans, leading to performance gains via resource, cost and information-processing efficiencies. In contrast, augmentation involves humans collaborating with machines, enhancing both human and machine skills

returns no significant direct effect of process automation on process performance once tasks become complex, and the recommendation they draw from it is a boundary condition: automate what is routine, augment what is ambiguous (Zebec & Indihar Štemberger, 2024, pp. 16–17)²⁰. Where the line falls is itself unstable. In the fragrance industry the machine took over part of the perfumers’ problem solving and could not touch the part that depends on smelling a scent and judging how people will react to it (Krakowski et al., 2023, p. 1445)²¹.

2.5 Situated AI

Kemp starts from the puzzle the two previous sections produce: AI is available to everyone, so buying it should not yield an advantage, yet firms plainly differ in what they get out of it. His answer is situated AI, which he defines as AI hemmed in by the firm’s own experience, structure and relationships (Kemp, 2024, p. 618)²². Three properties of the technology stand in the way, and the first of them is the imitability problem from Section 2.3 under another name: AI is generic, explicit and myopic. Being generic puts AI on a level with general human capital, which competitors can acquire and put to work much as the adopter does (Kemp, 2024, p. 620)²³.

What a firm does control is the situating. Kemp names three activities: grounding, which picks the experiences a firm’s own AI learns from; bounding, which works on the experiences feeding a competitor’s AI; and recasting, which keeps reworking the algorithms and the routines around them so the AI stays aligned with the rest of the firm’s activities (Kemp, 2024, pp. 618–619)²⁴. The vocabulary is conditional throughout. None of the three is a property of the technology, and each is something a firm may or may not do.

Empirical work in the corpus starts from the same reading. Shi et al. take the mixed results of earlier work as a sign that what AI yields depends on intangible complements, and productivity.”

²⁰[Zebec & Indihar Štemberger (2024), pp. 16–17] “The results show that BPA has no significant direct effect on BPP for more complex tasks in terms of efficiency of execution or scalability.” / “In summary, automate routine, well-structured tasks and augment complex, ambiguous ones.”

²¹[Krakowski et al. (2023), p. 1445] “While machines substitute perfumers regarding certain problem solving activities, they are unable to match the perfumers’ unique ability to smell fragrances and predict the human emotions they trigger. This example illustrates that AI is likely to substitute some, but not all of complex business tasks’ activities.”

²²[Kemp (2024), p. 618] “The present paper begins to resolve this puzzle by developing a theory of situated AI—AI whose agency is circumscribed in a firm’s experiential, structural, and relational systems.”

²³[Kemp (2024), p. 620] “The generic, explicit, and myopic nature of AI all endanger its firm-level benefits, but are uniquely difficult to surmount because AI is both a machine and agentic (Smith & Lewis, 2011). For example, AI’s generic nature is akin to general human capital. General human capital cannot typically underly interfirm advantages because a firm’s competitors can usually acquire and deploy that knowledge in ways that closely mimic corresponding actions of the focal firm.”

²⁴[Kemp (2024), pp. 618–619] “Grounding involves orchestrating which experiences one’s AI will be allowed to learn from across the organization. Bounding involves efforts to shape the experiences anchoring a competitor’s AI. Recasting involves orchestrating the continual adaptation of algorithms and their surrounding routines to enhance AI’s alignment with interdependent activities in a firm.”

on governance, and on managerial capability bound to its context, and they test that claim on Chinese listed firms, with data assets as the channel and managerial capability as the moderator (Shi et al., 2025, p. 1)²⁵. Kemp’s paper is conceptual, so what it offers are propositions rather than findings. They are the propositions this review is positioned to examine, because the conditions Kemp derives are the conditions the coded studies report.

²⁵[Shi et al. (2025), p. 1] “Drawing on a comprehensive dataset of Chinese listed companies spanning 2010–2022, this study investigates how AI adoption influences enterprise value through the mediating role of data assets and the moderating effect of managerial capability.” / “These mixed outcomes suggest that the value-creating capacity of AI depends on more than technological sophistication alone; it hinges on complementary intangible resources, effective governance structures, and context-specific managerial capabilities—factors that current empirical research has only partially explored.”

Chapter 3

Method

3.1 Review design

This thesis uses a systematic literature review to address its research question. The decision comes from the question itself: whether and under what conditions firm-level AI investments are profitable is an empirical issue on which data has rapidly accumulated across various outlets. This method is mostly chosen by researchers because of its comprehensiveness and transparency (Rojon et al., 2021, p. 196)¹. To allow for the reconstruction of each decision, every step that shapes the final corpus is recorded here.

The only data source is Scopus, which was accessed through the WU license. One recorded query is the source of every study that is included in the review, therefore all of the review’s conclusions are based entirely on the corpus. Other sources are only mentioned as methodology literature in this chapter and as theory anchors in Chapter 2, this review is restricted to peer-reviewed journal publications, no preprints or gray literature.

3.2 Search strategy

The search query was run on July 17 2026 through the Scopus Search API with the following query:

```
TITLE(("artificial intelligence" OR "machine learning" OR "deep
learning" OR "AI")) AND TITLE-ABS-KEY(("invest" OR "invests" OR
"invested" OR "investing" OR "investment" OR "investments" OR
"investor" OR "investors" OR adopt* OR implement*) AND ("competitive
advantage" OR "firm performance" OR "financial performance" OR "firm
productivity" OR "firm value" OR "market value" OR "firm growth")) AND
SUBJAREA(BUSI OR ECON OR DECI) AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND DOCTYPE(ar) AND LANGUAGE(english) AND SRCTYPE(j)
```

¹[Rojon et al. (2021), p. 196] “Scholars selected SRm for representing a rigorous, systematic, transparent and comprehensive approach to analysing, evaluating and synthesizing literature.”

The requirement that the AI terms have to appear in the title limits the search to studies where AI is the primary focus rather than a side topic. The investment and outcome vocabulary is searched more broadly, in title, abstract and keywords, because outcome measures are often named only in the abstract. Only terms are matched by Scopus, not concepts; a search phrase is not expanded to its synonyms, so a query looking simply for firm performance might overlook studies that refer to the same outcome as firm growth or firm value. Because a number of operations research publications publish firm-level AI evidence and are indexed there rather than under business or economics, decision sciences is included as a subject area. Additionally instead of only using the wildcard term **invest**, the investment terms are expressed word for word. An earlier version of the query found 75 additional records that only looked into AI without considering it as an investment, because the **invest** term also matches *investigate*.

A sound query must find Babina et al. (2024), Krakowski et al. (2023), and Lui et al. (2022), three studies that were set as retrieval benchmarks. The final query returns 432 different results including all three benchmark studies.

3.3 Selection criteria

What stays in the corpus is determined by two criteria.

The first is journal quality: a record is kept if its journal has a rating of at least 2 according to the Academic Journal Guide (AJG). The majority of the records that this rule eliminates came from journals that aren't even listed in the ranking at all. Every record, regardless of the year of publication, is subject to a single, fixed edition of the ranking, which maintains consistency and non-selectivity. The single source of truth for this thesis is the AJG 2024 edition.

The second layer has 4 criteria. A study has to be (1) empirical, (2) at the firm level, (3) focused on the firms' AI investment or adoption as its object of study, and (4) measuring a performance or competitive-advantage outcome. Only when all are met can the record stay in. Case studies are regarded as empirical and studies that survey managers stay in when the outcome is measured at the firm level.

In order to ensure that each excluded record has a single reason so the counts are clear, exclusions are assigned one of five codes in a fixed order with the first matching code gets assigned: **E1** not empirical (conceptual work, reviews, simulations without firm data); **E2** not at the firm level (individual, team, or country); **E3** AI is the study's analysis tool rather than the firms' investment object; **E4** no performance or competitive-advantage outcome (for example adoption intention or user satisfaction); **E5** wrong publication type. Across all criteria, a record that left doubt at the title and abstract stage was kept in for further review.

3.4 Screening process

Figure 3.1 shows the entire process. First the 432 records were reduced to 174 using the journal-quality filter, the record did not meet the minimum AJG ranking requirements. 93 of the 174 remaining records (E1: 20, E2: 7, E3: 39, E4: 27; E5 was never needed) were eliminated after the abstract and title were compared against the criteria. With the remaining 81 records the texts were then evaluated for eligibility using the same criteria, which were determined by an article’s method and results sections, another 14 more were excluded in this stage: 12 on the basis of unfitting content, and two whose texts were not available via open access or the university license. In total 67 studies are included in the review. As with the coding described below, screening was AI-assisted, with the author’s review and confirmation on every decision.

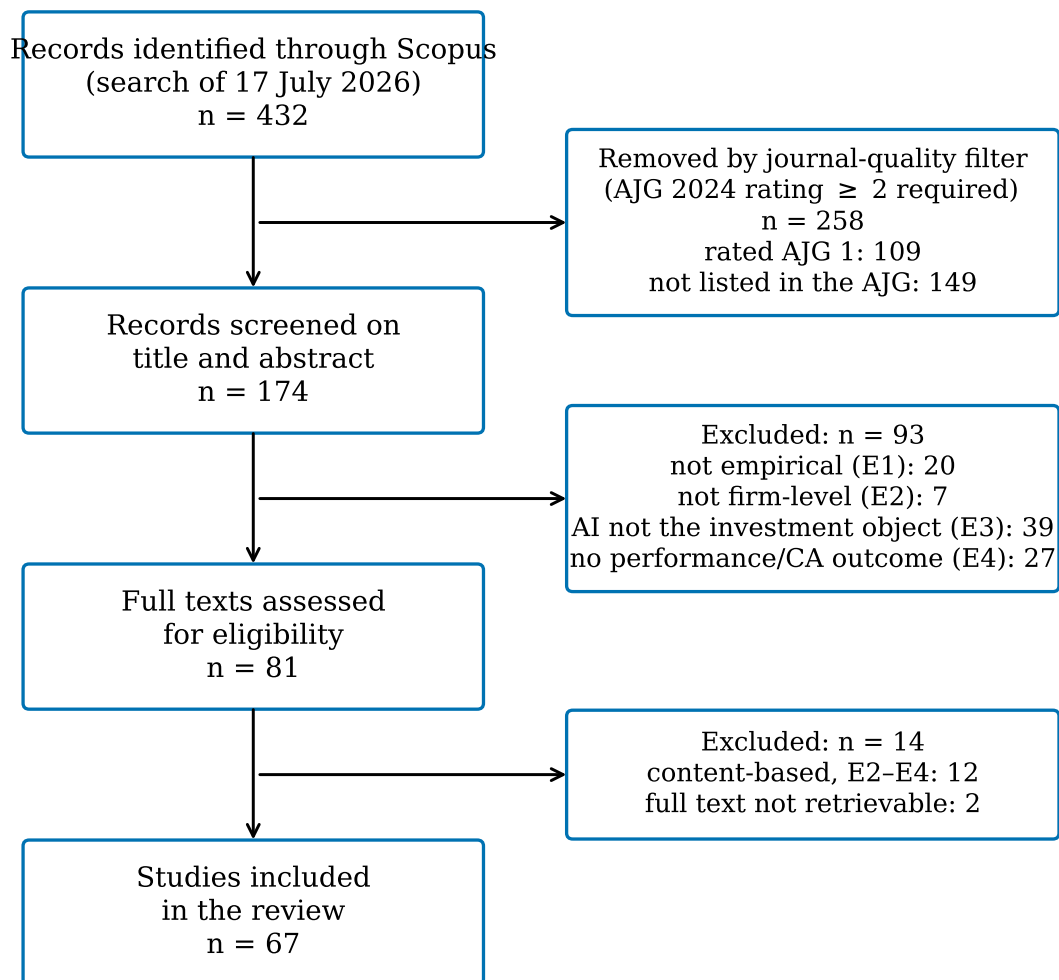


Figure 3.1: PRISMA-style flow of identification, screening, and inclusion. Source: own representation.

3.5 Coding

Each included study occupies one row of the coding table, split into study characteristics and findings. The scheme was fixed before coding began, with the fields adapted to the research question instead of adhering to a single template. The characteristics side documents how the evidence was produced, including its method, its sample and time frame, the country and industry it covers, the theoretical lens a study uses and how it measures AI investment, such as through surveys, AI-related job postings or adoption announcements.

The findings side records what the evidence says. It holds the outcome: performance, competitive advantage, or both. Performance means the firm improves by its own numbers, such as sales, profit, or productivity; competitive advantage means the firm does better than its competitors. Each construct has its own measurement field. The findings side records the direction of the effect, the identified conditions with their roles, a one-sentence key finding, and quality notes, for example when an outcome rests on managers' self-reports.

The core fields of this table are printed in the appendix (Tables A.1 and A.2). The theoretical lens, industry, and quality notes are incorporated into the thesis through the descriptive mapping and the discussion.

There are five possible options for the effect's direction. The average effect can be described as positive, negative, or null. Mixed means the finding splits across outcome measures, as when revenue rises but costs do not fall. Conditional means that the effect's size or sign relies on discovered conditions. Because they address different questions, direction indicates whether the investment pays on average, while conditions indicate what the payoff depends on. Direction and conditions are coded in different columns, as a result, a study might be classified as positive while still listing conditions. The conditions just enhance or reduce the effect on average.

Two general rules: A sample period or industry that is never mentioned in the publication remains null rather than being inferred since a cell only includes what a study reports. When a study's value falls outside of any of the categories, the cell contains the study's original text.

Each study is coded according to whether it measures competitive advantage, superior business performance, or both. Only when a study analyzes competitive advantage through a validated competitive-advantage scale, or using data comparing the company to its rivals does it qualify as measuring competitive advantage. Performance is coded as a performance gain alone. The first example is demonstrated by Chatterjee et al. (2021, p. 209)², who define competitive advantage in relation to other businesses in the same

²[Chatterjee et al. (2021), p. 209] "Competitive advantage is the degree to which the performance of an organization working in the B2B context could achieve greater benefits compared to other organizations functioning in similar conditions."

market. The second is demonstrated by Babina et al. (2024, p. 1)³: companies who invest in AI see faster growth in sales and employment. The study does not claim a position in relation to competitors, thus it is coded as performance. The table shows how rare it is for competitive advantages to be measured.

Every study was coded twice in independent, AI-assisted runs: Claude Fable 5 (Anthropic) drafted all rows as the first coder, and Gemini 2.5 Pro (Google) coded the same studies as the second, blind to the first run. Agreement between the runs was 90% for method, 88% for the outcome construct, and 73% for effect direction. The author checked every study's coded values against the results reported in the paper, including the studies on which the runs agreed, to confirm that the coding was justified. For the 29 studies on which the runs disagreed on at least one coded field, the author read the text and reached an independent decision before looking at either run's values, then resolved the disagreement against decision rules recorded in writing.

The conditions column also provides the structure of the synthesis in Section 4.2: every condition entry was assigned to one of six thematic groups, and a study stands in several groups when it identifies several conditions. The grouping followed the same dual principle as the coding.

³[Babina et al. (2024), p. 1] "AI-investing firms experience higher growth in sales, employment, and market valuations."

Chapter 4

Results

This chapter reports the evidence in two steps. Section 4.1 maps the 67 included studies descriptively: where the evidence comes from and how it is produced. Section 4.2 then synthesizes the conditions under which firm-level AI investments translate into performance gains or competitive advantage. All counts, groupings, and figures in this chapter are the author’s own compilation from the coding table; the full table is reported in the appendix.

4.1 Descriptive mapping of the evidence base

The search window ran from 2015 to 2026, yet the earliest included study dates from 2020 (Figure 4.1). Growth is concentrated at the end of that window: 44 of the 67 studies, two thirds of the corpus, appeared in 2025 and 2026 alone. The evidence base is young.

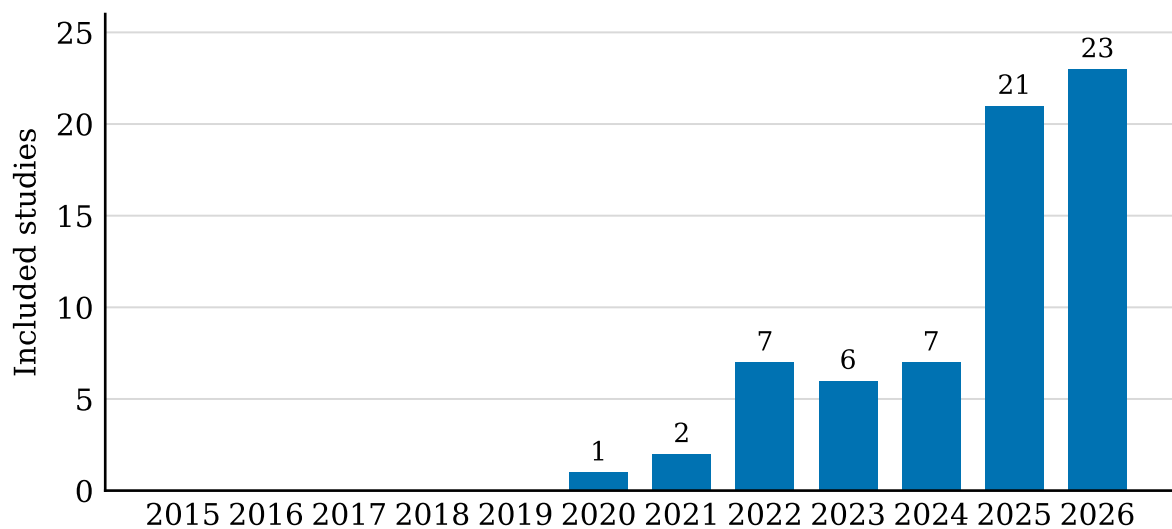


Figure 4.1: Included studies per publication year, search window 2015–2026 ($n = 67$). Source: own representation based on the coding table.

The 67 studies are spread across 45 journals, with no dominant outlet: *Industrial Marketing Management* and the *Journal of Business Research* lead with five studies each, followed by *Technology in Society*, the *International Review of Financial Analysis*, and the *International Journal of Information Management* with four each. Journal quality within the corpus is skewed toward the middle of the AJG scale (Figure 4.2): 27 studies come from journals rated 2, another 33 from journals rated 3, and seven from journals rated 4 or 4*.

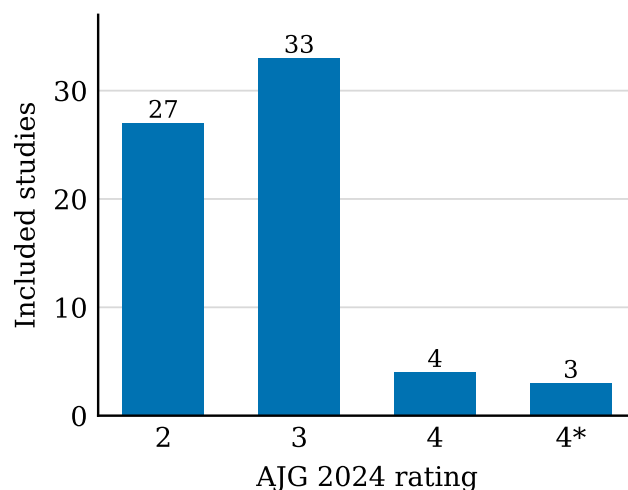


Figure 4.2: Included studies by AJG 2024 journal rating ($n = 67$). Source: own representation based on the coding table.

Two research designs dominate the corpus (Figure 4.3): panel econometrics on archival firm data (27 studies) and survey-based structural equation modeling (24 studies). The remaining sixteen studies combine several methods (six), examine stock market reactions in event studies (four), or build on case evidence (three); fsQCA, data envelopment analysis, and survey-based regression appear once each. The way studies operationalize AI investment splits the corpus in half: 33 studies rely on managers' self-reports through survey constructs, 31 on archival measures, and three on qualitative observation. Disclosure text mining (seven studies), patent-based proxies (six), workforce-based measures such as AI job postings (five), and investment announcements (four) account for most of the archival measures; the remaining nine rest on single-study instruments, for example procurement records or exposure to an AI policy shock.

Study samples cluster in two countries (Figure 4.4): the USA and China account for 16 studies each, together almost half of the corpus. India follows with seven studies, and another six draw on multi-country samples. European firms appear in eleven studies: five multi-country European samples plus six single-country ones. Most samples span several industries: 39 studies are cross-industry, ten focus on manufacturing, and the rest examine single settings such as healthcare or high-tech ventures.

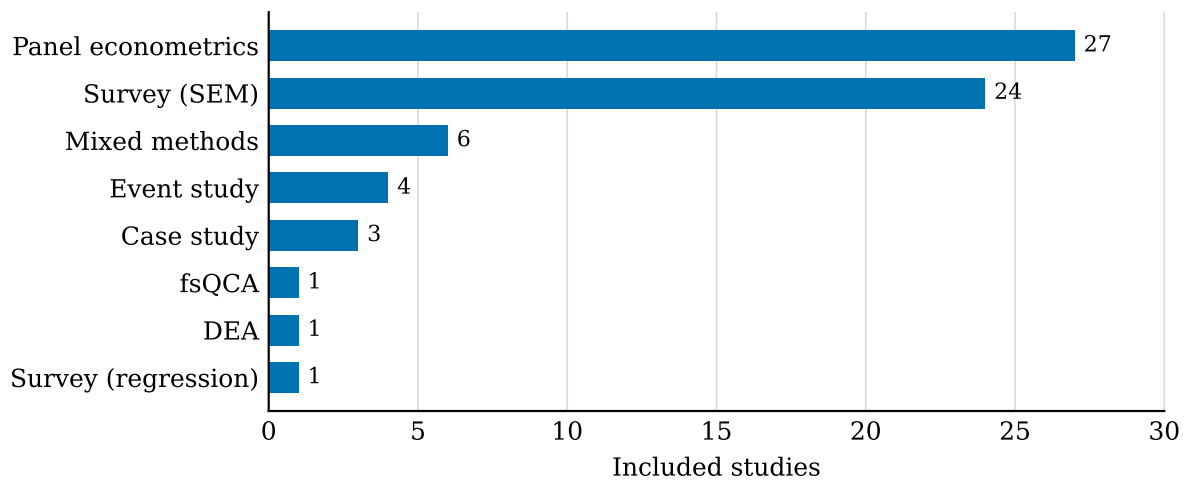


Figure 4.3: Included studies by research design ($n = 67$). Source: own representation based on the coding table.

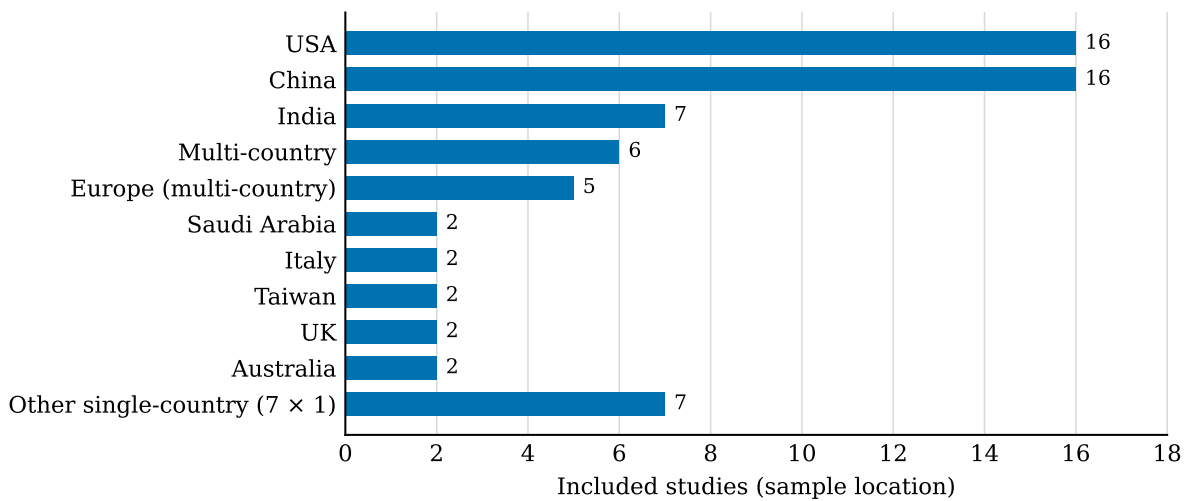


Figure 4.4: Included studies by sample location ($n = 67$). Source: own representation based on the coding table.

Following the distinction between superior firm performance and competitive advantage introduced in Chapter 2, the coding recorded which of the two constructs each study measures (Figure 4.5). The result is one-sided: 52 of the 67 studies measure firm performance only. Six measure competitive advantage as a distinct construct, and nine measure both. Measured competitive advantage is thus rare in this literature: 15 studies, under a quarter of the corpus. Effect directions spread across four of the five codes, and no study is coded null: 38 studies report a positive average effect of AI investment on their outcome, 19 an effect that exists only through a mechanism or under specific circumstances, six results that split across outcome indicators, and four negative effects. Among the 15 studies with a competitive-advantage construct, ten find positive and five conditional effects; none reports a negative one.

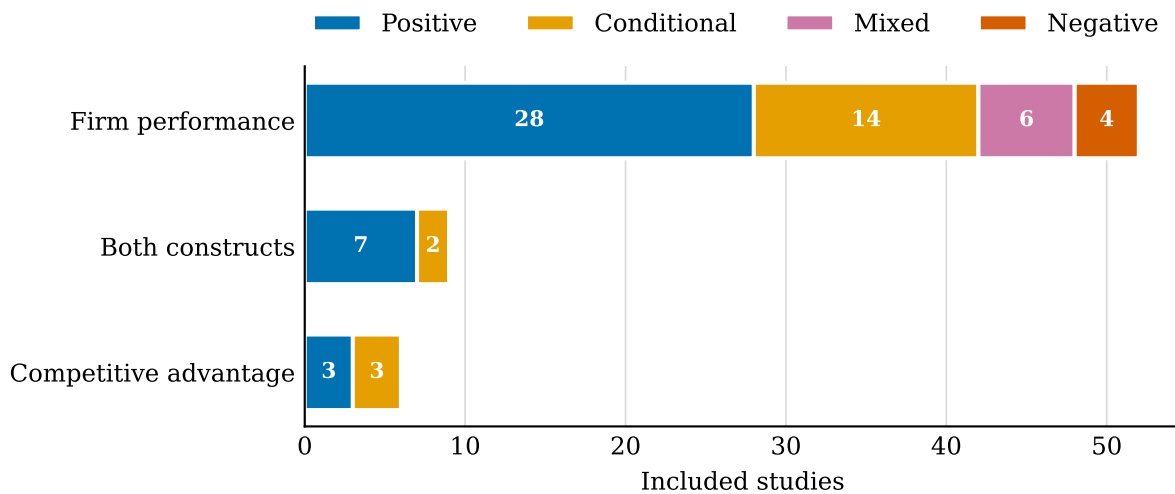


Figure 4.5: Outcome construct by reported effect direction ($n = 67$). Source: own representation based on the coding table.

Of the 67 studies, 63 identify at least one moderator, mediator, complement, or threshold that shapes the link between AI investment and its outcome. Only four report unconditional evidence. Section 4.2 orders these conditions.

4.2 Synthesis of conditions

Section 4.1 ended on the number that sets this section's task: 63 of the 67 studies identify at least one condition that shapes whether AI investment reaches its outcome. This section orders those conditions into six groups, moving from the investments a firm makes alongside AI to the timing and credibility of the investment itself. The grouping was fixed in a documented clustering of every condition entry in the coding table, and a study appears in several groups when it identifies several conditions. The four studies with unconditional evidence remain the contrast case.

4.2.1 Complementary investments and resources

Eleven of the 63 studies tie the payoff to what a firm invests alongside AI, and this group holds the most direct answers to the question of dose. Among high-tech ventures, revenue reacts only once AI adoption passes a sufficient intensity, and the gains grow where the venture also runs complementary technologies, cloud services and databases among them, and keeps R&D inside the venture (Lee et al., 2022, p. 1)¹. One panel study prices the threshold: the payoff to AI venture funding follows a U-shape whose turning point lies near USD 11.3 million, with short-term declines below that level and long-term gains above it (Banna & Alam, 2025, sec. 4.1)².

The complement named most often is the firm’s own research activity. In the same panel, AI investment alone stays insignificant while its interaction with R&D turns significantly positive (Banna & Alam, 2025, sec. 4.1)³. Evidence from UK firms points the same way: AI adoption raises innovation performance where it complements the firm’s internal R&D spending, and it partly substitutes collaboration with external knowledge partners (D’Amico et al., 2025, abstract)⁴.

Data holds the same position on the archival side. Among Chinese listed firms, data assets carry roughly 45.6 percent of the total value effect of AI adoption (Shi et al., 2025, p. 7)⁵. The case evidence describes that dependency from inside the organization: performance arrives only where firms rebuild their processes around the technology, and success rests on a bundle of data, talent, domain knowledge, partnerships and infrastructure (Wamba-Taguimdje et al., 2020, pp. 1893–1894)⁶.

¹[Lee et al. (2022), p. 1] “This paper examines how high-tech venture performance varies with AI-adoption intensity.” / “We find that firm revenue increases only after sufficient investment in AI, and the benefits of AI adoption are greater at firms that also invest in complementary technologies and pursue internal R&D strategy.” / “We find that such performance gains from adoption is larger among firms that invest in complementary technologies such as cloud computing and database systems.”

²[Banna & Alam (2025), sec. 4.1] “the turning point is observed at a log AI investment value of approximately 2.425 [...] corresponding to approximately USD 11.3 million in AI-related venture funding. This suggests that firms investing below this level may face short term performance declines, while those exceeding this threshold begin to realise the long-term benefits of AI adoption”

³[Banna & Alam (2025), sec. 4.1] “AI investment alone has a positive but statistically insignificant relationship with revenue growth, while R&D alone shows a negative association.” / “the interaction term [...] exhibits a positive and statistically significant effect, indicating that the combination of AI adoption and R&D activity produces complementary and mutually reinforcing effects on firm performance.”

⁴[D’Amico et al. (2025), abstract] “Using data from 14,143 firms in the UK between 2004 and 2020, we explore how AI adoption impacts knowledge spillover of innovation.” / “Our findings suggest that firms adopting AI can enhance their innovation performance if it complements firm’s own investment in R&D. AI adoption can substitute knowledge collaboration with certain external partners.”

⁵[Shi et al. (2025), pp. 1, 7] “Drawing on a comprehensive dataset of Chinese listed companies spanning 2010–2022, this study investigates how AI adoption influences enterprise value through the mediating role of data assets and the moderating effect of managerial capability.” (p. 1) / “The comparable magnitudes of direct (0.0483) and indirect (0.0405) effects suggest that data assets mediate approximately 45.6% of AI adoption’s total effect on firm value, highlighting the substantial role of data assets in translating AI capabilities into enterprise value.”

⁶[Wamba-Taguimdje et al. (2020), pp. 1893–1894] “The same results also showed that organizations achieve performance through AI capabilities only when they use their features/technologies to reconfigure their processes.” / “[...] must be brought together to ensure the success of AI: data, talent mix, domain

The survey studies phrase the same requirement as readiness. Whether Saudi B2B SMEs take up AI practices depends on technology roadmapping, attitude, awareness and infrastructure; professional expertise and technicality stay insignificant (Baabdullah et al., 2021, p. 255)⁷. The study behind it treats technological readiness and ICT capability as entry conditions for AI-supported partner management, and both significantly improve its performance (Samadhiya et al., 2023, p. 510)⁸. In recruitment, adoption reaches competitive advantage through dynamic capabilities, built from HR competencies together with open innovation; financial backing and IT infrastructure support these capabilities (Sandeep et al., 2025, p. 404)⁹.

Two studies extend the complement logic to how the technology is sourced and combined. French firms that source AI externally face more volatile productivity outcomes than in-house developers, and the volatility fades where ICT specialists make up more of the payroll (Fontanelli et al., 2025, pp. 1–2)¹⁰. Pairing AI with robotic process automation adds revenue, mainly through prices, but does not cut costs further, and the revenue gain concentrates where the firm’s strategic intent targets growth (Arshad et al., 2026, p. 1)¹¹. The case studies read the same list from its negative side: data quality, missing skills, high investment needs and unclear economic benefits are the barriers that stand between AI and supply-chain competitiveness (Cannas et al., 2024, p. 3333)¹². Across the eleven

knowledge, key decisions, external partnerships and scalable infrastructure.”

⁷[Baabdullah et al. (2021), p. 255] “this study empirically examines antecedents to, and consequences of, successful acceptance of AI practices by B2B SMEs in Saudi Arabia.” / “The results showed that, of the AI enablers, acceptance of AI practices was significantly influenced by both technology roadmapping and attitude but not professional expertise. Of the AI readiness variables, acceptance of AI practices was significantly influenced by infrastructure and awareness but not technicality.”

⁸[Samadhiya et al. (2023), p. 510] “[...] including information communication technology capability, firm fit, and technological readiness, as fundamental requirements for the implementation of AI-based partner relationship management.” / “The findings of the study show that information communication technology capability and technological readiness play a significant role in improving the performance of AI-based partner relationship management.”

⁹[Sandeep et al. (2025), p. 404] “This study examines the influence of organizational resources on AI adoption in recruitment, focusing on their role in achieving competitive advantage through effective implementation.” / “The results reveal that HR competencies and open innovation significantly influence dynamic capabilities, which are essential for AI integration, supported by financial support and information technology (IT) infrastructure.” / “These capabilities enable effective AI adoption, leading to a competitive advantage.”

¹⁰[Fontanelli et al. (2025), pp. 1–2] “This paper shifts the focus and investigates the impact of predictive AI on the volatility of firms’ productivity growth rates. Using firm-level data from the 2019 French ICT survey, we provide robust evidence that AI use is associated with increased volatility.” / “Our results show that heightened volatility is concentrated among AI buyers, whereas firms that develop AI internally experience no such association.” / “we find that the AI-volatility link among ‘AI buyers’ is mitigated in firms with a higher share of ICT engineers and technicians, suggesting that AI’s successful integration requires complementary human capital.”

¹¹[Arshad et al. (2026), p. 1] “We find that joint implementation is positively associated with additional revenue growth (linked primarily to price increases) but is not associated with additional cost reduction beyond the separate effects of RPA and AI.” / “among firms implementing both RPA and AI, a stronger revenue growth intent is associated with greater revenue-related outcomes.”

¹²[Cannas et al. (2024), p. 3333] “The results highlighted how AI methods in OSCM can increase the companies’ competitiveness by reducing costs and lead times and improving service levels, quality, safety, and sustainability.” / “However, they also identify barriers in the implementation of AI, such as ensuring

studies of this group, a pattern emerges: none reports that acquiring the technology alone is enough.

4.2.2 Organizational capabilities and human capital as the channel

With 26 of the 63 studies, this is the largest group, and its shared claim is structural: the investment does not act on performance directly, it builds an organizational capability, and the capability delivers the outcome. The claim is strongest where the direct path is empty. Among early-stage ventures, strategic resilience carries the entire link between AI capability and entrepreneurial success, and the authors read the mechanism as organizational, not technological (Wang & Zhang, 2026, p. 1)¹³. Italian manufacturers show the same shape: AI reaches manufacturing firm performance through knowledge management processes (Leoni et al., 2022, p. 411)¹⁴. Even value co-creation follows the pattern: its impact on competitive advantage stays insignificant, and the firms’ innovation and dynamic capabilities do the transmitting (Y. Sun et al., 2022, p. 474)¹⁵. Adoption lifts organisational performance by way of the quality of decisions and the performance of business processes, with process innovation, automation and organisational learning as partial, complementary mediators (Zebec & Indihar Štemberger, 2024, p. 1)¹⁶. One two-stage survey models AI capabilities as inputs to the firm’s adaptive response to market change and finds that this response lifts firm performance and innovation (Sullivan & Fosso Wamba, 2024, p. 1)¹⁷. In accounting firms, AI-infused knowledge pays through green human and structural capital, while the relational channel contributes nothing (Bin-Nashwan & Li,

data quality, lack of specific skills, need for high investments, lack of clarity on economic benefits and lack of experience in cost analysis for AI projects.”

¹³[Wang & Zhang (2026), p. 1] “we collected two-wave survey data from 357 early-stage ventures in China and Europe” / “The mediating effect of strategic resilience is robust and fully accounts for the AI capability–success relationship, indicating that the mechanism through which AI investments generate performance returns is fundamentally organizational rather than technological in nature.”

¹⁴[Leoni et al. (2022), p. 411] “six hypotheses have been developed and tested through an empirical survey administered to 120 senior executives of Italian manufacturing firms.” / “the authors show the positive effects of the adoption of AI on KMPs, as well as the influence of KMPs on SCR and MFP. Finally, the authors demonstrate that KMPs act as a mediator through which AI affects SCR and MFP.”

¹⁵[Sun et al. (2022), p. 474] “impacts of economic value and relationship value on competitive advantage and innovation intelligibility are not significant; and (3) value co-creation does not have a significant impact on competitive advantage and innovation intelligibility, and dynamic capabilities and innovation capabilities play the mediating role.”

¹⁶[Zebec & Indihar Štemberger (2024), p. 1] “AI adoption increases organisational performance through decision-making and business process performance. Process automation, organisational learning and process innovation are significant complementary partial mediators”

¹⁷[Sullivan & Fosso Wamba (2024), p. 1] “Our research employs a two-stage design, surveying IT and business executives from firms that have adopted AI.” / “This research examines how AI-powered capabilities can bring value to organizations by enhancing their Adaptive Response to Market Changes (ARMC).” / “we find that ARMC positively influences firm performance and innovation.”

2025, p. 1)¹⁸. And big-data intelligence raises new product performance by being converted into AI capabilities, with electronic supply-chain collaboration strengthening the indirect path (Tao et al., 2025, abstract)¹⁹.

The dynamic-capabilities studies specify what the channel is made of. Garment exporters hold sustained competitive advantage through sensing the market, seizing it and reconfiguring it, and all three run higher once AI adoption rests on an analytics foundation (Hossain et al., 2022, p. 240)²⁰. In Malaysian hospitality, sustainable competitive advantage transmits adoption into value, while dynamic capabilities hold value back during the transition, a drag that technological turbulence deepens rather than relieves (Alam et al., 2025, pp. 1, 9)²¹. Among manufacturing SMEs in emerging economies, AI knowledge capabilities work directly and through more radical digitally enabled product innovation, with digital sustainability leadership reshaping both paths (Dinh, 2026, p. 84)²². In Chinese manufacturing, AI capabilities raise sustainability performance via a tiered mechanism whose prescriptive component fails (Renfei et al., 2026, p. 1)²³.

A cluster of survey studies locates the channel in the marketing function. For food firms, a marketing strategy built on AI raises performance only inside configurations, combined with customer co-creation, marketing capabilities and market orientation (Wu & Monfort, 2023, p. 484)²⁴. For Chinese cross-border e-commerce SMEs, the pairing of

¹⁸[Bin-Nashwan & Li (2025), p. 1] “AIK positively and significantly affected the internal dimensions of GIC, green human capital (GHC) and green structural capital (GSC) within accounting firms, while green rational capital (GRC) had no effect.”

¹⁹[Tao et al. (2025), abstract] “BDAI assimilation significantly promotes NPP, and AI capabilities act as an intermediary bridge – empowering B2B firms to convert BDAI assimilation into enhanced NPP.” / “This study further found that this mediation model is strengthened through the contingency impact of ESCC”

²⁰[Hossain et al. (2022), p. 240] “This research investigates how industrial goods manufacturers sustain their competitive advantage in export markets, convincing buyers in a competitive data-rich business environment. The evidence has been taken from the RMG (readymade garment) industry” / “the study reveals that firms marketing analytics capability play a vital role in sensing, seizing, and reconfiguring the market, consequently leading to a sustained competitive advantage. The performance of sensing, seizing, and reconfiguring becomes higher for a firm when they adopt AI on the strength of the marketing analytics platform.”

²¹[Alam et al. (2025), pp. 1, 9] “Sustainable competitive advantage emerged as a significant mediator, translating AI adoption and dynamic capabilities into enhanced value creation.” (p. 1) / “Conversely, dynamic capability (DC) shows a negative relationship with value creation ($\beta = -0.385$, $p < 0.001$)” (p. 9) / “Technological turbulence (TT) moderates these relationships, particularly weakening the positive impact of dynamic capabilities on value creation ($\beta = -0.140$, $p=0.025$).” (p. 9)

²²[Dinh (2026), p. 84] “Emerging market manufacturing small and medium-sized enterprises (SMEs) struggle to translate artificial intelligence (AI) investments into innovation outcomes.” / “AI-enabled knowledge capabilities operate through dual pathways: directly enhancing performance and indirectly through digital product innovation radicalness.” / “this study identifies digital sustainability leadership as a critical boundary condition acting as both mediator and moderator, fundamentally reconfiguring how AI generates value.”

²³[Renfei et al. (2026), p. 1] “This study empirically analyzes data from Chinese manufacturing enterprises through the PLS-SEM method.” / “AIC significantly enhances CSP through a tiered enabling mechanism, but prescriptive capabilities fail to achieve their intended effects.”

²⁴[Wu & Monfort (2023), p. 484] “Data were collected from 278 food firms.” / “The research results of FsQCA find that causal conditions of marketing capabilities, customer value co-creation, market orientation, and AI marketing strategy are necessary and sufficient recipes for higher firm performance.”

AI adoption with strategic agility strengthens product innovation as well as the upgrading of brands through international marketing capabilities (Guo et al., 2025, p. 1250)²⁵. Stakeholder engagement built on AI reaches firm performance through marketing agility (Tehrani & Roy, 2025, abstract)²⁶. CRM systems powered by AI pay through marketing ambidexterity; profitability and competitive advantage follow (Alnofeli et al., 2026, p. 1)²⁷. For generative AI, technological opportunism converts into competitive advantage through three affordances (Li, Wu, et al., 2026, p. 1)²⁸.

The human side of the channel is its own line of evidence. The benchmark study of this group states it sharpest: in chess, AI adoption makes the players' traditional strengths obsolete and creates capabilities at the human-machine boundary that have no relation, at times a negative one, to the old ones (Krakowski et al., 2023, p. 1425)²⁹. Cohesive teams improve how knowledge is shared, and a committed workforce widens the performance gain from adoption (Ullah et al., 2026, p. 1)³⁰. AI human capital raises firm value partly by compressing wage inequality (Agag et al., 2026, p. 1)³¹. AI capabilities reach sustainable SME performance through green innovation and creativity (Alwakid & Dahri, 2025, p. 1)³². The archival studies find organizational channels as well. Among Chinese listed companies, the key route from AI to productivity is a drop in information asymmetry; more specialized work division and green innovation appear as potential channels (Z. Sun et al., 2024, p. 1)³³. And the retail evidence turns the channel configurational: performance comes

²⁵[Guo et al. (2025), p. 1250] "A quantitative approach was employed to analyse data from 322 small and medium-sized enterprises in China utilizing cross-border platforms." / "the coupling of AI adoption and strategic agility positively influences IMCs, product innovation and brand upgrading, while IMCs serve as a mediating mechanism."

²⁶[Tehrani & Roy (2025), abstract] "The results indicate that engaging stakeholders through AI technologies positively influences marketing agility, which in turn enhances firm performance. These effects are amplified in conditions of high technology turbulence."

²⁷[Alnofeli et al. (2026), p. 1] "The quantitative analysis confirms that AI-powered CRM capabilities positively shape marketing ambidexterity, which subsequently enhances profitability and competitive advantage."

²⁸[Li et al. (2026), p. 1] "find that three GenAI affordances—organizational memory, collaborative, and creation affordances—mediate the relationship between TO and competitive advantages."

²⁹[Krakowski et al. (2023), p. 1425] "AI adoption triggers interrelated substitution and complementation dynamics, which make humans' traditional competitive capabilities obsolete, while creating new sources of persistent heterogeneity when humans interact with chess engines. These novel human-machine capabilities are unrelated, or even negatively related, to traditional capabilities."

³⁰[Ullah et al. (2026), p. 1] "cohesive team dynamics enhance collaboration and knowledge sharing" / "a highly committed workforce amplifies the performance benefits of AI adoption by fostering deeper employee engagement, greater adaptability, and enhanced organizational flexibility."

³¹[Agag et al. (2026), p. 1] "Findings indicate that AIHC is positively associated with firm value, negatively associated with wage inequality, and that wage inequality is negatively associated with firm value." / "Mediation analyses suggest that wage inequality partially channels the AIHC and firm value association."

³²[Alwakid & Dahri (2025), p. 1] "The results indicate that AI capabilities significantly impact SME creativity and green innovations. Moreover, green entrepreneurial orientation positively influences SME creativity and green innovations, which in turn facilitate sustainable performance."

³³[Sun et al. (2024), p. 1] "Based on panel data of 3235 listed companies in China from 2007 to 2021, we comprehensively discuss the impact and mechanism of AI on firm productivity" / "Mitigating information asymmetry is a key channel, while specialized division of labor and independent green innovation are

from combinations of AI applications rather than single uses, and out of the combinations grow new capabilities, adaptive learning among them (Cao et al., 2025, p. 1)³⁴. Taken together, the studies of this group agree on the order: the investment builds a capability, the capability earns the return, and several find no significant direct effect at all.

4.2.3 Leadership, governance, and management

Thirteen of the 63 studies condition the payoff on who runs the firm and how it is governed. On the survey side the recurring moderator is leadership. In AI-embedded CRM, leadership support lifts both relationship satisfaction and firm performance (Chatterjee et al., 2022, p. 437)³⁵. The companion study reports the same moderating role of leadership support for firm performance under AI-CRM (Chatterjee et al., 2021, p. 213)³⁶. Generative AI adoption lifts firm performance, and ethical leadership moderates that effect (Kumar et al., 2025, p. 1)³⁷. In hotels and resorts, leader narcissism shapes how the adoption factors translate into competitive positioning (Chakraborty, 2025, abstract)³⁸. The take-up of AI by key account managers is shaped by their personality traits, and organizational culture moderates the agreeableness path (Mehta et al., 2025, p. 543)³⁹. On the archival side, innovation gains from AI are amplified where executives hold IT backgrounds (Feng et al., 2025, p. 1)⁴⁰.

Managerial capability marks the boundary of the effect in the panel evidence. Firms with a stronger management team pull more value out of the investment (Shi et al., 2025, potential ones.”

³⁴[Cao et al. (2025), p. 1] “we first construct a validated measurement instrument based on grounded analysis of 6,519 AI-related documents from 37 retailers. We then apply fuzzy-set qualitative comparative analysis (fsQCA) to survey data from 140 executives” / “superior performance stems not from isolated AI uses but from synergistic combinations—particularly those involving customer service and cybersecurity—interacting with other functions. These configurations appear to give rise to emergent capabilities such as adaptive learning, predictive analytics, and uncertainty mitigation”

³⁵[Chatterjee et al. (2022), p. 437] “the study highlights that there is a negative impact of the moderator ‘technology turbulence’ on the relations of ‘automated decision making’ and ‘operational efficiency’ with ‘B2B relationship satisfaction’, whereas there is a positive impact of moderator ‘leadership support’ on ‘B2B relationship satisfaction’ and ‘firm performance’.”

³⁶[Chatterjee et al. (2021), p. 213] “our study has shown that for exhibiting better performance, leadership support has a major moderating effect on the firm performance in the B2B context using an AI-CRM system”

³⁷[Kumar et al. (2025), p. 1] “The results also highlight that GenAI adoption boosts firm performance. Finally, ethical leadership was found to moderate the effect of GenAI adoption on firm performance.”

³⁸[Chakraborty (2025), abstract] “This study examines the use of Generative AI (GAI) in the hospitality industry, particularly in hotels and resorts.” / “integrating the Technology-Organization-Environment (T-O-E) framework alongside leadership traits such as narcissism.” / “this study further explains how narcissism moderates the relationship between the factors that assist in adopting GAI and GAI’s moderating effect on the overall competitive positioning and organisational outcomes.”

³⁹[Mehta et al. (2025), p. 543] “Our research shows that key account managers’ adoption of AI technologies is influenced significantly by personality traits.” / “Organisational culture significantly moderates the association between agreeableness and the adoption of AI technologies.”

⁴⁰[Feng et al. (2025), p. 1] “Further analysis shows that AI-driven innovation is amplified in firms with executives who have information technology backgrounds, and that highly innovative firms diversify their supply chain to reduce resource risk.”

p. 1)⁴¹. One efficiency study pushes the point furthest: the baseline association between AI investment and efficiency is zero, and gains appear mainly where capable managers, sharper competitive pressure, stable institutional ownership and long-horizon financing come together (Kazakis, 2026, abstract)⁴².

The remaining studies move from persons to governance systems. Among Chinese listed companies, part of the AI–TFP link runs through improved control quality (M. Lin et al., 2026, abstract)⁴³. Investment efficiency gains from AI are more pronounced where knowledge intensity and R&D spending are high and the internal controls sound (N. Lin et al., 2026, p. 1)⁴⁴. Responsible AI reaches competitive advantage through supply-chain justice, and complexity erodes the distributive channel while the procedural one holds (Li, Shi, et al., 2026, p. 1)⁴⁵. The qualitative pole reports the same dependency: AI-adopting pharmaceutical companies manage tendering faster and decide better, yet run into governance and compliance challenges, and the payoff hinges on aligning adoption with the EU’s AI Act and the GDPR (Pesqueira et al., 2024, p. 1)⁴⁶. In this group’s reading, the same investment returns more under better management and tighter governance.

4.2.4 Firm characteristics and market position

Sixteen of the 63 studies find the payoff sorted by what the firm already is, and size leads the list. In the corpus’s benchmark result, growth from AI investment concentrates

⁴¹[Shi et al. (2025), p. 1] “The findings establish that managerial capability positively moderates this relationship, indicating that firms with stronger management teams extract greater value from AI investments.”

⁴²[Kazakis (2026), abstract] “Using a labor-based measure of AI investment, the baseline results show no direct association between AI investment and firm efficiency. The heterogeneity analysis indicates that efficiency gains materialize primarily when firms pair AI with capable managers, face stronger competitive pressure, have more stable institutional investor ownership, and are able to secure long-term debt.”

⁴³[Lin, Chang et al. (2026), abstract] “This research investigates the impact of AI adoption on total factor productivity (TFP) within Chinese A-share listed companies from 2007 to 2022.” / “we identify a key transmission mechanism: the positive relationship between AI and TFP is partially channeled through enhanced internal control quality (ICQ).”

⁴⁴[Lin, Li et al. (2026), p. 1] “The results show that AI can significantly enhance the investment efficiency of enterprises. This effect is more pronounced in high knowledge-intensity firms, high R&D-intensity firms, and firms with sound internal controls.”

⁴⁵[Li, Shi et al. (2026), p. 1] “Survey data from 218 Chinese firms show that responsible AI fosters both distributive and procedural justice, which in turn facilitates firms’ competitive advantages. Further, we identify an asymmetric moderating effect: high supply chain complexity weakens the mediating effect of distributive but not procedural justice.”

⁴⁶[Pesqueira et al. (2024), p. 1] “This research examines the incorporation of artificial intelligence (AI) into the domain of tender management (TM) within the pharmaceutical industry, with a particular emphasis on operational efficiency, governance, and compliance with European regulatory standards.” / “The findings indicate that companies integrating AI demonstrate enhanced decision-making capabilities, accelerated processing times, and enhanced stakeholder engagement. However, they also encounter challenges pertaining to ethical governance and regulatory compliance.” / “The research highlights the necessity of aligning the adoption of AI with the latest European directives, such as the AI Act and General Data Protection Regulation (GDPR), to ensure both operational efficiency and adherence to ethical standards.”

among the larger firms; industry concentration rises with it (Babina et al., 2024, p. 1)⁴⁷. AI-powered pricing shows the same order: the bigger, more productive players are the likelier adopters, and adopters grow faster in sales and employment as well as markups (Adams et al., 2026, p. 1)⁴⁸. Hospitals holding more market share are the likelier adopters and improve revenue, occupancy and productivity with AI, with the authors stressing that assessing these gains requires endogeneity controls (Pham et al., 2024, p. 1)⁴⁹. Innovation gains cluster in big incumbents and firms in their growth stage (Feng et al., 2025, p. 1)⁵⁰. The reverse also holds: when AI services fail, small firms lose more market value than large ones, and accuracy failures cost the most (D. Song et al., 2025, p. 504)⁵¹.

Ownership and financial position sort the effect as well. The productivity gain is largest in state-controlled firms, in internationally active ones, and in innovators (Z. Sun et al., 2024, p. 1)⁵². The valuation premium favors state ownership, vanishes in polluting industries unless AI belongs to a green transition that is credible, and appears only where asset structures stay light (Shi et al., 2025, p. 1)⁵³. Acquisitions after adoption concentrate in firms holding abundant cash, in less developed market environments, and under private ownership (D. Liu et al., 2026, p. 1)⁵⁴. Efficiency gains need, next to capable managers, stable institutional ownership and long-horizon financing (Kazakis, 2026, abstract)⁵⁵. Investment efficiency responds more where knowledge intensity, R&D spending

⁴⁷[Babina et al. (2024), p. 1] “AI-powered growth concentrates among larger firms and is associated with higher industry concentration.”

⁴⁸[Adams et al. (2026), p. 1] “At the firm level, larger and more productive firms are more likely to adopt AI pricing. Firms that adopted AI pricing experienced faster growth in sales, employment, assets, and markups, and their stock returns are also more responsive to high-frequency monetary policy surprises than non-adopters.”

⁴⁹[Pham et al. (2024), p. 1] “we find that hospitals with a larger market share are great candidates to adopt AI. Furthermore, these hospitals can leverage AI technology to enhance various aspects of performance, including total outpatient revenue, total inpatient revenue, productivity, and occupancy. Importantly, we demonstrate that controlling for endogeneity is essential in assessing the performance outcomes of AI adoption.”

⁵⁰[Feng et al. (2025), p. 1] “Heterogeneity analysis reveals that large incumbents and growth-stage firms leverage AI most effectively for innovation outputs and efficiency.”

⁵¹[Song et al. (2025), p. 504] “First, AI service failure events have a negative impact on the firm value. Second, small firms experience more share price declines due to AI service failure events than large firms.” / “there are order effects on firm value across the service failure event types: accuracy > safety > privacy > fairness.”

⁵²[Sun et al. (2024), p. 1] “AI significantly improves firm productivity, especially in state-controlled, internationally minded, and innovative firms.”

⁵³[Shi et al. (2025), p. 1] “Three heterogeneity tests reveal boundary conditions: the valuation premium is stronger for state-owned enterprises, disappears in heavily polluting industries unless AI is part of a credible green transition, and is evident only in non-asset-intensive firms where resource reconfiguration is agile.”

⁵⁴[Liu et al. (2026), p. 1] “We find that AI adoption significantly increases a firm’s likelihood of pursuing M&A, with the effect especially pronounced among firms with abundant cash holdings and those operating in less-developed market environments. The relationship is also stronger for privately owned enterprises, and AI adoption is associated with improved long-term firm value.”

⁵⁵[Kazakis (2026), abstract] “The heterogeneity analysis indicates that efficiency gains materialize primarily when firms pair AI with capable managers, face stronger competitive pressure, have more stable institutional investor ownership, and are able to secure long-term debt.”

and internal controls are strong (N. Lin et al., 2026, p. 1)⁵⁶. And the market’s reward for AI innovation splits along firm traits: return on assets shows a small positive effect at best, stronger for firms heavy on R&D, while firm value rises where sustainability metrics are good, with Democratic administrations amplifying the reward (Singh et al., 2026, abstract)⁵⁷.

The downside sorts by the same variables. Investor reactions to AI investment announcements are negative on average, a drop of 1.77 percent on the announcement day, and nonmanufacturing firms, those with a weak IT base, and those with poor credit ratings take the harder hit (Lui et al., 2022, p. 373)⁵⁸. Crash risk after AI investment grows where information transparency is low and resources are tight (Bai et al., 2026, p. 1)⁵⁹. In Taiwanese semiconductors, innovation from AI lifts sustainability efficiency, while profit efficiency tracks size together with leverage (Chiu et al., 2025, abstract)⁶⁰. Industry-4.0 manufacturers convert AI capabilities into sustainability performance far better than traditional ones (Renfei et al., 2026, p. 1)⁶¹. The value association of AI human capital is stronger in travel, the compression of wage inequality stronger in hospitality (Agag et al., 2026, p. 1)⁶². Across this group, most sorting runs in one direction: firms that are larger, better resourced, or better rated capture more of the upside and less of the downside.

4.2.5 Environmental conditions

Fifteen of the 63 studies place the condition outside the firm. On the survey side the recurring variable is technological turbulence, and it cuts both ways. It dampens the paths from automated decision making and operational efficiency to relationship satisfaction

⁵⁶[Lin, Li et al. (2026), p. 1] “This effect is more pronounced in high knowledge-intensity firms, high R&D-intensity firms, and firms with sound internal controls.”

⁵⁷[Singh et al. (2026), abstract] “The study finds mixed evidence of a small positive impact of AI innovation on firms’ return on assets, with the effect stronger for more R&D-intensive firms.” / “The findings demonstrate that AI innovation increases firm value for companies that perform well on sustainability-related metrics. This effect is stronger under Democratic administrations than under Republican governments”

⁵⁸[Lui et al. (2022), p. 373] “The stock prices of the firms decrease by 1.77% on the day of the announcement. Nonmanufacturing firms and firms with weak information technology capabilities or low credit ratings suffer a more negative impact compared with other firms.”

⁵⁹[Bai et al. (2026), p. 1] “we find robust evidence that AI investment significantly increases firms’ stock price crash risk, mainly due to reduced information transparency and heightened managerial optimism. The effect is more pronounced for firms with lower levels of information transparency and tighter resource constraints.”

⁶⁰[Chiu et al. (2025), abstract] “applying a three-stage Data Envelopment Analysis (DEA) to evaluate the performance of 33 publicly listed semiconductor firms in Taiwan across profitability, sustainability, and market value” / “Tobit regression analysis shows that AI-driven innovation enhances sustainability efficiency, whereas firm size and leverage are more closely linked to profitability.”

⁶¹[Renfei et al. (2026), p. 1] “The impact of AIC on CSP is significantly weaker in traditional manufacturing firms than in Industry 4.0 manufacturing firms.”

⁶²[Agag et al. (2026), p. 1] “Findings indicate that AIHC is positively associated with firm value, negatively associated with wage inequality, and that wage inequality is negatively associated with firm value.” / “Sectoral analyses reveal stronger value associations in travel and stronger inequality compression associations in hospitality.”

(Chatterjee et al., 2022, p. 437)⁶³. In hospitality it moderates selectively, touching only the path from dynamic capabilities to value (Alam et al., 2025, p. 1)⁶⁴. On the other side it amplifies: the payoff to AI-based stakeholder engagement grows when technology turbulence is high (Tehrani & Roy, 2025, abstract)⁶⁵. Hostile and fast-moving environments condition how much each AI capability feeds the firm’s adaptive response (Sullivan & Fosso Wamba, 2024, p. 1)⁶⁶. Competition sharpens the payoff as well. Competitive intensity strengthens the creational path from technological opportunism to competitive advantage (Li, Wu, et al., 2026, p. 1)⁶⁷. The efficiency gains of the corpus’s null-baseline study need, among other conditions, sharper competitive pressure (Kazakis, 2026, abstract)⁶⁸.

Regulation can raise or lower the payoff. Where the regulatory climate is clear and its enforcement supportive, the link from AI-enabled innovation to productivity strengthens (Z. Liu et al., 2026, p. 1)⁶⁹. Exposure to privacy regulation blunts the gain in investment efficiency (Chen et al., 2026, p. 1)⁷⁰. The pharmaceutical tender case declares alignment with the EU’s AI Act and the GDPR necessary for the payoff (Pesqueira et al., 2024, p. 1)⁷¹. Dubai’s robotic-surgery advantage required the health authority’s approval one

⁶³[Chatterjee et al. (2022), p. 437] “the study highlights that there is a negative impact of the moderator ‘technology turbulence’ on the relations of ‘automated decision making’ and ‘operational efficiency’ with ‘B2B relationship satisfaction’, whereas there is a positive impact of moderator ‘leadership support’ on ‘B2B relationship satisfaction’ and ‘firm performance’.”

⁶⁴[Alam et al. (2025), p. 1] “This study explores the relationship between technology readiness, AI adoption, dynamic capabilities, sustainable competitive advantage, and technological turbulence, assessing their collective impact on value creation within the Malaysian hospitality industry.” / “the moderating effects of technological turbulence were selective, significantly influencing only the relationship between dynamic capabilities and value creation.”

⁶⁵[Tehrani & Roy (2025), abstract] “These effects are amplified in conditions of high technology turbulence.”

⁶⁶[Sullivan & Fosso Wamba (2024), p. 1] “The results demonstrate significant interaction effects of environmental hostility and dynamism on the relationships between AI-powered capabilities and ARMC.”

⁶⁷[Li, Wu et al. (2026), p. 1] “we ground in affordance theory to explore how firms can leverage GenAI affordances to transform technological opportunism (TO) into sustained competitive advantages, particularly in highly competitive environments.” / “competitive intensity amplifies the mediation effect of creational affordance instead of the mediation effects of organizational memory and collaborative affordances.”

⁶⁸[Kazakis (2026), abstract] “Using a labor-based measure of AI investment, the baseline results show no direct association between AI investment and firm efficiency.” / “The heterogeneity analysis indicates that efficiency gains materialize primarily when firms pair AI with capable managers, face stronger competitive pressure, have more stable institutional investor ownership, and are able to secure long-term debt.”

⁶⁹[Liu, Weng & Yu (2026), p. 1] “Results suggest that AI-enabled innovation is associated with firm growth both directly and indirectly through productivity gains, with the strength of the innovation–productivity link contingent on the regional regulatory climate.” / “Regulatory clarity and supportive enforcement are manipulated, and results indicate that clear and supportive environments are associated with higher levels of intended AI investment, productivity expectations, deployment speed, and growth projections.” / “moderated mediation findings reveal that the productivity pathway is strongest when both clarity and supportiveness are present.”

⁷⁰[Chen et al. (2026), p. 1] “Our baseline results indicate that higher AI adoption is associated with greater investment efficiency” / “Cross-sectional evidence further indicates that regulatory frictions, such as exposure to data-privacy regulation, attenuate the association between AI adoption and investment efficiency.”

⁷¹[Pesqueira et al. (2024), p. 1] “This research examines the incorporation of artificial intelligence

case at a time, next to a shift in medical culture (Ali Mohamad et al., 2023, pp. 53, 61)⁷². Even partisan politics enters: the market’s reward for sustainability-conscious AI innovation grows under Democratic administrations (Singh et al., 2026, abstract)⁷³.

The macro environment sorts the effect too. The growth payoff of AI investment, 0.04 percent per ten-percent increase in investment, hangs on the labour market: labour productivity works for it, labour costs and the labour share work against it (Tingbani et al., 2025, p. 961)⁷⁴. Trade tensions push suppliers into AI integration, and the integration improves efficiency, sales and the firm’s market value until extreme exposure to tariffs overwhelms the compensation (Filieri et al., 2026, p. 57)⁷⁵. Policy can push the other way: digital-transformation policies along the supply chain add to the productivity gain (Z. Sun et al., 2024, p. 1)⁷⁶. Stock market reactions to AI-heavy service patents in manufacturing are negative, softened by idiosyncratic volatility (Patel, 2026, p. 1)⁷⁷. What unites this group: the payoff shifts with the environment even when the investment does not.

4.2.6 Dose, credibility, and timing

The last group, fourteen of the 63 studies, conditions the payoff on how much is invested, how credibly it is communicated, and when. Three of its dose findings stand in earlier groups already: the adoption-intensity threshold and the U-shaped funding payoff belong to the complements group (Section 4.2.1), the tariff turning point to the environment

(AI) into the domain of tender management (TM) within the pharmaceutical industry, with a particular emphasis on operational efficiency, governance, and compliance with European regulatory standards.” / “The research highlights the necessity of aligning the adoption of AI with the latest European directives, such as the AI Act and General Data Protection Regulation (GDPR), to ensure both operational efficiency and adherence to ethical standards.”

⁷²[Ali Mohamad et al. (2023), pp. 53, 61] “introducing breakthrough technology in the health sector, such as Artificial Intelligence, necessitates a cultural shift inside the medical community” (p. 53) / “the numerous advantages and benefits generated by the adoption of a robotic approach led the DHA (Dubai Health Authority) to approve the treatment of automated surgery cases in Dubai on a case-by-case basis, depending on the complexity and necessity.” (p. 61)

⁷³[Singh et al. (2026), abstract] “The findings demonstrate that AI innovation increases firm value for companies that perform well on sustainability-related metrics. This effect is stronger under Democratic administrations than under Republican governments”

⁷⁴[Tingbani et al. (2025), p. 961] “Using the generalized method of moments (GMM) estimation on 1950 unique American firms over 1996–2016, we show that a 10% increase in AI investment leads to an increase in firm growth by 0.04%. However, this result is highly sensitive to labour market conditions, as labour productivity can positively impact firm growth, but labour cost and labour share negatively influence firm growth.”

⁷⁵[Filieri et al. (2026), p. 57] “Results indicate that higher tariff exposure increases AI integration, particularly among firms with stronger customer engagement needs. AI integration improves supply chain efficiency, sales performance, and market value. However, performance benefits follow an inverted U-shape: extreme tariff exposure diminishes AI’s compensatory effects.”

⁷⁶[Sun et al. (2024), p. 1] “Supply chain digital transformation policies enhance the productivity effect of AI”

⁷⁷[Patel (2026), p. 1] “In a sample of 1306 service technology patents and 758,143 non-service technology patents from 3899 manufacturing firms (1980–2020)” / “stock market reactions to service technology patents with high AI scores are negative” / “we find that idiosyncratic volatility mitigates this negative reaction, while tangibility shows no such effect.”

group (Section 4.2.5). What this group adds is the shape of the middle: platform embeddedness helps most there, with medium depth doing the most for product innovation and moderate breadth the most for brand upgrading (Guo et al., 2025, p. 1250)⁷⁸.

How the investment is deployed carries its own conditions. The case evidence ties performance to firms reworking their processes around AI (Wamba-Taguimdje et al., 2020, p. 1893)⁷⁹. Externally bought AI comes with a heightened volatility that in-house development does not show (Fontanelli et al., 2025, pp. 1–2)⁸⁰. Pairing AI with robotic process automation adds revenue rather than cost savings, and the gain follows the strategic intent behind the deployment (Arshad et al., 2026, p. 1)⁸¹.

Credibility decides the market’s reaction. Announcements of AI implementation earn a positive reaction on the day itself, and the returns differ with the detail of the announcement (Huang & Lee, 2023, p. 9)⁸². In corporate filings, actionable disclosures with clear plans bring valuation gains while speculative and irrelevant mentions bring nothing, and silence or vagueness draws penalties (Basnet et al., 2025, p. 1)⁸³. Unsubstantiated claims, the practice the SEC labels AI washing, first attract investors and then turn into negative reactions and lasting underperformance (X. Song et al., 2026, p. 1)⁸⁴.

Time itself conditions the estimates. The productivity effect of AI loses pace over the long run (Z. Sun et al., 2024, p. 1)⁸⁵. The negative market impact of AI service failures has grown more intense in recent years (D. Song et al., 2025, p. 504)⁸⁶. The negative

⁷⁸[Guo et al. (2025), p. 1250] “platform embedding characteristics exhibit a nonlinear, inverted U-shaped moderating effect: a medium level of embedding depth maximizes the coupling’s positive impact on product innovation, whereas a moderate level of embedding breadth optimizes its positive effect on brand upgrading.”

⁷⁹[Wamba-Taguimdje et al. (2020), p. 1893] “The same results also showed that organizations achieve performance through AI capabilities only when they use their features/technologies to reconfigure their processes.”

⁸⁰[Fontanelli et al. (2025), pp. 1–2] “Our results show that heightened volatility is concentrated among AI buyers, whereas firms that develop AI internally experience no such association.”

⁸¹[Arshad et al. (2026), p. 1] “We find that joint implementation is positively associated with additional revenue growth (linked primarily to price increases) but is not associated with additional cost reduction beyond the separate effects of RPA and AI.” / “among firms implementing both RPA and AI, a stronger revenue growth intent is associated with greater revenue-related outcomes.”

⁸²[Huang & Lee (2023), p. 9] “in response to the announcement of AI implementation, we find significant positive abnormal returns on the event day.” / “the results show significant differences in average cumulative abnormal returns between announcements with detailed information and those without detailed information.”

⁸³[Basnet et al. (2025), p. 1] “Actionable disclosures outlining clear implementation plans yield significant valuation benefits, particularly upon first introduction, whereas speculative or irrelevant disclosures have no impact.” / “early adopters of actionable disclosures gain competitive advantages, while peers that either remain silent or offer only vague AI disclosures face market penalties.”

⁸⁴[Song, Hou et al. (2026), p. 1] “In March 2024, SEC Chair Gary Gensler labeled this practice ‘AI washing.’” / “we find that such narratives initially attract investors but ultimately yield negative market reactions and sustained underperformance.” / “markets ultimately punish rhetoric that outpaces implementation.”

⁸⁵[Sun et al. (2024), p. 1] “the dynamic decomposition effect shows that the productivity-enhancing effect of AI is slowing down in the long run.”

⁸⁶[Song et al. (2025), p. 504] “Third, AI service failure events in more recent years have a more intensively negative impact than those in more distant years.”

reaction to AI-heavy service patents shows only in the two most recent decades (Patel, 2026, p. 1)⁸⁷. Market value and financial performance rise with implementation, but the edge of first movers and top performers is not uniform across the indicators (Huang & Lin, 2025, p. 1856)⁸⁸. Across this group, dose, credibility, and timing each move the payoff.

Four studies of the corpus report unconditional evidence; the other 63 tie the payoff to at least one of the conditions ordered in this section.

⁸⁷[Patel (2026), p. 1] “stock market reactions to service technology patents with high AI scores are negative” / “Decade-wise analysis reveals effects are significant only in the 2010s and 2020s, suggesting evolving market perceptions of AI-driven servitization.”

⁸⁸[Huang & Lin (2025), p. 1856] “this research reveals that while AI adoption enhances financial performance and market value, the advantages for AI first movers and better performers are not uniformly positive across all indicators.”

Chapter 5

Discussion

This review set out to answer one question: under what conditions do firm-level AI investments translate into competitive advantage and superior firm performance? The preceding chapter delivered the material for the answer; this chapter interprets it.

5.1 The conditional answer

The first half of the answer concerns performance, and it is positive but conditioned. 38 of the 67 studies report a positive average effect, against 19 conditional, six mixed and four negative ones. The positive count alone would overstate the case: 63 of the 67 studies, the positive ones included, identify at least one condition that shapes whether the investment reaches its outcome, and only four report evidence without any. The six groups of Section 4.2 can be read as one sequence: a firm invests alongside the technology, converts the investment into organizational capabilities, steers the conversion through leadership and governance, within what the firm already is and where it operates, at a dose and with a credibility the market accepts. This ordering is the review’s own synthesis; no single study tests it end to end.

The conditions rarely arrive alone. In the corpus’s null-baseline study, efficiency gains appear where capable managers, competitive pressure, stable institutional ownership and long-horizon financing come together (Kazakis, 2026, abstract)¹. The configurational evidence makes the combination itself the unit of analysis. For food firms, performance follows recipes that join marketing capabilities, customer co-creation, market orientation and an AI marketing strategy (Wu & Monfort, 2023, p. 484)². In retail, the payoff grows

¹[Kazakis (2026), abstract] “Using a labor-based measure of AI investment, the baseline results show no direct association between AI investment and firm efficiency. The heterogeneity analysis indicates that efficiency gains materialize primarily when firms pair AI with capable managers, face stronger competitive pressure, have more stable institutional investor ownership, and are able to secure long-term debt.”

²[Wu & Monfort (2023), p. 484] “Data were collected from 278 food firms.” / “The research results of FsQCA find that causal conditions of marketing capabilities, customer value co-creation, market orientation, and AI marketing strategy are necessary and sufficient recipes for higher firm performance.”

out of combinations of AI applications rather than any single use (Cao et al., 2025, p. 1)³. Read this way, the practical question is not which condition matters most. It is which configurations a firm can complete.

The second half of the answer concerns competitive advantage, and it is mostly missing. 15 of the 67 studies measure the construct at all: ten report positive evidence, five conditional. The remaining 52 measure performance and stop there. Barney’s test for a sustained advantage, survival after rivals’ attempts at duplication have run their course (Barney, 1991, p. 102)⁴, is not met by survey scales that capture a perceived edge at one point in time. Across the 15, this review reads only the chess evidence of Krakowski et al. as a test of persistence: the capabilities that emerge at the human-machine boundary stay heterogeneous over time (Krakowski et al., 2023, p. 1425)⁵. Performance gains are documented. Sustained competitive advantage is measured rarely, and almost never as persistence.

5.2 Reading the evidence through the four lenses

The general-purpose-technology lens predicted an uneven payoff driven by complements, and the corpus delivers its firm-level picture. Revenue reacts only past an adoption-intensity threshold and grows where complementary technologies and internal R&D stand behind the investment (Lee et al., 2022, p. 1)⁶. The funding payoff turns at roughly USD 11.3 million (Banna & Alam, 2025, sec. 4.1)⁷. Behind both stands the intangible logic of the productivity J-curve: complements are built, unbooked, before they pay (Brynjolfsson et al., 2021, pp. 333–334)⁸. The lag that Brynjolfsson et al. place in the national accounts

³[Cao et al. (2025), p. 1] “we first construct a validated measurement instrument based on grounded analysis of 6,519 AI-related documents from 37 retailers. We then apply fuzzy-set qualitative comparative analysis (fsQCA) to survey data from 140 executives” / “superior performance stems not from isolated AI uses but from synergistic combinations—particularly those involving customer service and cybersecurity—interacting with other functions.”

⁴[Barney (1991), p. 102] “Rather, whether or not a competitive advantage is sustained depends upon the possibility of competitive duplication. Following Lippman and Rumelt (1982) and Rumelt (1984), a competitive advantage is sustained only if it continues to exist after efforts to duplicate that advantage have ceased.”

⁵[Krakowski et al. (2023), p. 1425] “AI adoption triggers interrelated substitution and complementation dynamics, which make humans’ traditional competitive capabilities obsolete, while creating new sources of persistent heterogeneity when humans interact with chess engines.”

⁶[Lee et al. (2022), p. 1] “We find that firm revenue increases only after sufficient investment in AI, and the benefits of AI adoption are greater at firms that also invest in complementary technologies and pursue internal R&D strategy.”

⁷[Banna & Alam (2025), sec. 4.1] “the turning point is observed at a log AI investment value of approximately 2.425 [...] corresponding to approximately USD 11.3 million in AI-related venture funding. This suggests that firms investing below this level may face short term performance declines, while those exceeding this threshold begin to realise the long-term benefits of AI adoption”

⁸[Brynjolfsson et al. (2021), pp. 333–334] “realizing their potential also requires large intangible investments and a fundamental rethinking of the organization of production itself. Firms must create new business processes, develop managerial experience, train workers, patch software, and build other intangibles. This raises productivity measurement issues because intangible investments are not readily

shows up here as thresholds in firm panels. The lens holds.

The resource-based view holds up against the evidence in its strict reading only. A purchasable technology meets none of Barney’s four conditions (Section 2.3), and the corpus behaves accordingly: several studies that model a direct path from investment to outcome find it insignificant (Section 4.2.2). Wang and Zhang state the resulting reading most directly: the mechanism behind AI’s performance returns is organizational, not technological (Wang & Zhang, 2026, p. 1)⁹. Krakowski et al. place the imitation barrier in the bundle of human and machine capabilities rather than in either part (Krakowski et al., 2023, p. 1446)¹⁰. The firm-characteristics group adds the sorting the RBV expects. Growth from AI concentrates among the larger firms (Babina et al., 2024, p. 1)¹¹. Weakness sorts the downside the same way: firms with weak IT bases or poor credit ratings take the deeper announcement penalty (Lui et al., 2022, p. 373)¹². What the RBV cannot do is carry the 52 performance-only studies: higher margins show value creation, not inimitability. The construct separation this thesis maintains marks exactly the difference between what the literature has shown and what it has not.

The automation-augmentation distinction enters the evidence as a design condition. Zebec and Indihar Štemberger find no significant direct effect of process automation on process performance once tasks grow complex, and derive the boundary rule: automate the routine, augment the ambiguous (Zebec & Indihar Štemberger, 2024, pp. 16–17)¹³. Sourcing adds a corollary: externally bought AI carries a volatility that in-house development does not show (Fontanelli et al., 2025, pp. 1–2)¹⁴. Raisch and Krakowski’s argument that the two applications cannot be kept apart in practice reads, after this corpus, like a job description for management. The moderators of Section 4.2.3, from leadership support to managerial capability, describe who draws that line: firms under stronger management

tallied on a balance sheet or in the national accounts.” / [p. 334] “The error in measured total factor productivity growth therefore follows a J-curve shape, initially dipping while the investment rate in unmeasured capital is larger than the investment rate in other types of capital, then rising as growing intangible stocks begin to contribute to measured production.”

⁹[Wang & Zhang (2026), p. 1] “The mediating effect of strategic resilience is robust and fully accounts for the AI capability–success relationship, indicating that the mechanism through which AI investments generate performance returns is fundamentally organizational rather than technological in nature.”

¹⁰[Krakowski et al. (2023), p. 1446] “Our findings show that humans remain sources of sustainable advantage by combining and integrating their capabilities with machine capabilities into uniquely complementary bundles (Argyres & Zenger, 2012).” / “the locus of sustainable advantage is neither human nor machine capabilities, but what humans do with these capabilities.”

¹¹[Babina et al. (2024), p. 1] “AI-powered growth concentrates among larger firms and is associated with higher industry concentration.”

¹²[Lui et al. (2022), p. 373] “Nonmanufacturing firms and firms with weak information technology capabilities or low credit ratings suffer a more negative impact compared with other firms.”

¹³[Zebec & Indihar Štemberger (2024), pp. 16–17] “The results show that BPA has no significant direct effect on BPP for more complex tasks in terms of efficiency of execution or scalability.” / “In summary, automate routine, well-structured tasks and augment complex, ambiguous ones.”

¹⁴[Fontanelli et al. (2025), pp. 1–2] “Our results show that heightened volatility is concentrated among AI buyers, whereas firms that develop AI internally experience no such association.”

pull more value out of the same investment (Shi et al., 2025, p. 1)¹⁵.

Kemp’s situated AI fits the overall pattern best, and it is the least tested of the four lenses. The theory holds that a generic technology yields nothing defensible by itself, because rivals can buy and deploy the same thing (Kemp, 2024, p. 620)¹⁶. It gives the firm three activities to work against that: grounding, bounding and recasting (Kemp, 2024, pp. 618–619)¹⁷. The conditionality this corpus documents is what such a theory would predict. The six condition groups can be read as the empirical surface of that claim, with complements and capabilities on the grounding side and governance on the recasting side; this mapping is the review’s own proposal, not something the studies test. Direct tests are beginning just outside the corpus. On 570 Spanish manufacturing firms, Vaillant et al. find that firms using AI platforms show more cooperative and capability-building behavior when identifying problems and implementing solutions (Vaillant et al., 2025, abstract)¹⁸. That cooperation leads toward the grounding, bounding and recasting capabilities the theory names, in what the authors call one of the first empirical applications of situated AI (Vaillant et al., 2025, secs. 6.1, 6.2)¹⁹. That study measures no performance outcome and was excluded from this corpus for exactly that reason. As a template for testing Kemp’s propositions against outcomes, it is the more valuable.

5.3 Implications for practice

For a firm weighing the investment, the corpus prices the entry ticket. Below the turning point of roughly USD 11.3 million in AI venture funding, performance can decline in the short run (Banna & Alam, 2025, sec. 4.1)²⁰, and the complements that decide the payoff, from data assets to specialists to reworked processes, cost money the technology budget

¹⁵[Shi et al. (2025), p. 1] “The findings establish that managerial capability positively moderates this relationship, indicating that firms with stronger management teams extract greater value from AI investments.”

¹⁶[Kemp (2024), p. 620] “For example, AI’s generic nature is akin to general human capital. General human capital cannot typically underly interfirm advantages because a firm’s competitors can usually acquire and deploy that knowledge in ways that closely mimic corresponding actions of the focal firm.”

¹⁷[Kemp (2024), pp. 618–619] “Grounding involves orchestrating which experiences one’s AI will be allowed to learn from across the organization. Bounding involves efforts to shape the experiences anchoring a competitor’s AI. Recasting involves orchestrating the continual adaptation of algorithms and their surrounding routines to enhance AI’s alignment with interdependent activities in a firm.”

¹⁸[Vaillant et al. (2025), abstract] “Data collected from 570 Spanish manufacturing firms in 2023 reveals that firms utilizing AI platforms exhibit greater cooperative and situating capability-building behavior during the problem identification and solution implementation stages.”

¹⁹[Vaillant et al. (2025), sec. 6.1] “It was found that such usage does indeed facilitate cooperation during the problem identification and solution implementation stages, and, as predicted by the theory of situated AI, this cooperation leads solution ecosystems towards greater AI-driven situating capabilities, namely, those attained through grounding, bounding, and recasting.” / [sec. 6.2] “this is one of the first studies to empirically apply the situated AI theory.”

²⁰[Banna & Alam (2025), sec. 4.1] “the turning point is observed at a log AI investment value of approximately 2.425 [...] corresponding to approximately USD 11.3 million in AI-related venture funding. This suggests that firms investing below this level may face short term performance declines, while those exceeding this threshold begin to realise the long-term benefits of AI adoption”

does not show. The downside is also not symmetric: when AI services fail, small firms lose more market value than large ones (D. Song et al., 2025, p. 504)²¹. A firm that cannot fund the complements has, on this evidence, little reason to expect the average positive effect.

Management is a working condition, not a background variable. Across the governance group, the same investment returns more under leadership support, capable management and sound internal controls (Section 4.2.3); none of these arrive with the software. The capability channel of Section 4.2.2 adds the sequencing: the investment builds a capability first and earns its return second, which means the payoff arrives only after the organizational work is done.

Communication about the investment carries payoff conditions of its own. Actionable disclosure with clear plans earns valuation gains while speculative mentions earn nothing (Basnet et al., 2025, p. 1)²². Rhetoric that outruns implementation is punished: AI washing draws investors in at first, and what follows are negative market reactions and lasting underperformance (X. Song et al., 2026, p. 1)²³.

5.4 Limitations

Four limitations bound these conclusions. The first is composition: 33 of the 67 studies measure AI through survey constructs and self-reports, and many of their outcomes are perceived rather than accounted. The second is identification, and most of the corpus is associational. The hospital evidence makes the point from inside the corpus: assessing AI’s payoff requires endogeneity controls (Pham et al., 2024, p. 1)²⁴. The third is coverage: samples from the United States and China make up 47.8 percent of the corpus, and two thirds of the studies (65.7 percent) appeared in 2025 and 2026, so long-run outcomes are largely unobservable, and the one study that decomposes the effect over time finds it slowing already (Z. Sun et al., 2024, p. 1)²⁵. The fourth is the review protocol itself: a single database, journal articles in English, a journal-quality threshold of AJG 2, and a query scoped to investment rather than any AI use, all documented in Chapter 3. A corpus assembled this way can only contain what journals publish, and null results are

²¹[Song et al. (2025), p. 504] “Second, small firms experience more share price declines due to AI service failure events than large firms.”

²²[Basnet et al. (2025), p. 1] “Actionable disclosures outlining clear implementation plans yield significant valuation benefits, particularly upon first introduction, whereas speculative or irrelevant disclosures have no impact.”

²³[Song, Hou et al. (2026), p. 1] “we find that such narratives initially attract investors but ultimately yield negative market reactions and sustained underperformance.” / “markets ultimately punish rhetoric that outpaces implementation.”

²⁴[Pham et al. (2024), p. 1] “we find that hospitals with a larger market share are great candidates to adopt AI.” / “Importantly, we demonstrate that controlling for endogeneity is essential in assessing the performance outcomes of AI adoption.”

²⁵[Sun et al. (2024), p. 1] “the dynamic decomposition effect shows that the productivity-enhancing effect of AI is slowing down in the long run.”

the finding least likely to reach publication; the four-study unconditional contrast group should be read with that in mind.

5.5 Future research

The largest gap follows from the construct separation: competitive advantage needs to be measured as persistence, not as a perceived edge. 15 of 67 studies measure the construct, and almost all of them capture it cross-sectionally; designs that follow adopters and their imitators over years, in the way the chess panel follows players, would show whether any AI-based edge survives duplication. Kemp’s theory offers the propositions and now has a first operationalization to build on; connecting Vaillant et al.’s situating capabilities to performance and persistence outcomes is the obvious next step. The dose findings deserve replication outside venture funding, since a threshold that moves with firm size or industry would change the practical advice. The make-or-buy margin, so far documented once for volatility, deserves the same attention for performance levels. Coverage should widen beyond the two economies that dominate the samples. And the corpus’s newest layer, generative AI, is still measured at adoption; its performance evidence has yet to accumulate.

Chapter 6

Conclusion

This thesis asked under what conditions firm-level AI investments translate into competitive advantage and superior firm performance, and answered it with a systematic review of 67 empirical studies published between 2020 and 2026, drawn from a documented Scopus search of 432 records, screened in three recorded stages, and coded by two independent coders against a frozen scheme. The coding table in the appendix is the review's core deliverable; every number in the text derives from it.

The answer favors conditions over the investment itself. Across the corpus, 38 of the 67 studies report positive average effects, and the evidence is conditional almost throughout: 63 studies identify at least one condition that decides whether the investment reaches its outcome. The review ordered these conditions into six groups, from the complementary investments a firm makes alongside the technology to the credibility with which the investment is communicated. For competitive advantage, the evidence is thin. 15 studies measure the construct, almost always as a perceived edge at one point in time, and only a single study in the corpus follows an advantage against imitation over time.

The contribution is the ordered map itself. Where individual studies report single moderators and mediators, the review shows that the conditions recur in patterned groups across methods and settings, that the configurational evidence finds them working in combination rather than alone, and that the two outcome constructs the field often blends carry different amounts of evidence. The map marks where the next studies are needed, with persistence the least covered and most consequential claim. Firms can read the same map as a checklist of what has to be in place before the investment can be expected to pay.

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Appendix

Coding table

The full coding table for all 67 included studies, split into study characteristics (Table A.1) and findings (Table A.2).

Table A.1: Included studies: characteristics and design (n = 67). Source: author's compilation.

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S01	Wamba-Taguimdje et al. (2020)	Business Process Management Journal (2)	multi-country	~500 vendor-published AI project cases	case study	case observation (vendor project reports)
S02	Baabdullah et al. (2021)	Industrial Marketing Management (3)	Saudi Arabia	392 B2B SMEs	survey-SEM	survey construct (acceptance of AI practices)
S03	Chatterjee et al. (2021)	Industrial Marketing Management (3)	India	349 managers	survey-SEM	survey construct (AI-CRM implementation)
S04	Chatterjee et al. (2022)	Journal of Business Research (3)	India	312 manager responses	survey-SEM	survey construct (AI-embedded CRM adoption)
S05	Hossain et al. (2022)	Industrial Marketing Management (3)	Bangladesh	257 manager responses	survey-SEM	survey construct (AI adoption on a marketing analytics platform)
S06	Lee et al. (2022)	Technovation (3)	South Korea	160 high-tech firms	panel econometrics	survey (AI-adoption intensity) linked to financial records
S07	Leoni et al. (2022)	International Journal of Operations and Production Management (4)	Italy	120 senior executives	survey-SEM	survey construct (AI adoption/use)
S08	Lui et al. (2022)	Annals of Operations Research (3)	USA	119 AI investment announcements by 62 listed firms, 2015-2019	event study	announcements (AI investment announcements)
S09	Mishra et al. (2022)	Journal of the Academy of Marketing Science (4*)	USA	19,000 firm-year observations of US-listed firms	panel econometrics	annual-report text (share of AI-related keywords)
S10	Sun et al. (2022)	Systems Research and Behavioral Science (2)	China	234 AI-related manufacturers (general-manager/CEO respondents)	survey-SEM	survey construct (value co-creation in AI innovation ecosystem)
S11	Ali et al. (2023)	Journal of Organizational Change Management (2)	UAE	single hospital case (Dubai): 9 interviews with the robotic-surgery team plus archival data	case study	case observation (adoption of AI/robotic surgery)
S12	Czarnitzki et al. (2023)	Journal of Economic Behavior and Organization (3)	Germany	5,851 firms (409 AI users), representative German innovation survey 2018	panel econometrics	survey (AI adoption yes/no plus intensity of AI use in business processes)
S13	Huang & Lee (2023)	Data Base for Advances in Information Systems (2)	USA	109 AI implementation announcements by S&P 500 firms, 2014-2019	event study	announcements (AI implementation)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S14	Krakowski et al. (2023)	Strategic Management Journal (4*)	multi-country	same chess players across human-only, human-plus-AI and engine-only tournaments (controlled competitive setting)	panel econometrics	adoption of chess engines (AI) in competitive play
S15	Samadhiya et al. (2023)	Industrial Marketing Management (3)	India	interviews with senior B2B managers plus a survey of 142 responses	mixed	survey construct (AIPRM capabilities: AI-based partner relationship management)
S16	Wu & Monfort (2023)	Psychology and Marketing (3)	Taiwan	278 CEO/marketing-manager responses, food franchises and chain stores, 2020	survey-SEM	survey construct (implementation of AI marketing strategy)
S17	Babina et al. (2024)	Journal of Financial Economics (4*)	USA	1,993 US listed firms	panel econometrics	resume-based (share of AI-skilled workers as a proxy for firm AI investment)
S18	Cannas et al. (2024)	International Journal of Production Research (3)	Italy	6 companies, 17 AI implementation cases, interview-based	case study	case observation (AI applications across supply-chain processes)
S19	Pesqueira et al. (2024)	Computers and Industrial Engineering (2)	Europe	4 pharmaceutical companies (2 AI adopters vs. 2 non-adopters) plus a two-round expert panel of 53	mixed	adoption status (AI in tender management)
S20	Pham et al. (2024)	Journal of Business Research (3)	USA	941 AI and 941 matched non-AI hospital-year observations (246 hospitals), US, 2000-2020	panel econometrics	adoption dummy (hospital has adopted AI or not)
S21	Sullivan & Fosso (2024)	Journal of Business Research (3)	Europe	107 executives (France + UK), two-wave survey	survey-SEM	survey constructs (AI-enabled automation, AI-enabled analytics, AI-enabled relational capabilities)
S22	Sun et al. (2024)	Technology in Society (2)	China	3,235 Chinese listed firms, 2007-2021	panel econometrics	annual-report text (frequency of AI-related terms)
S23	Zebec & Indihar (2024)	Business Process Management Journal (2)	EU	448 organisations, EU	survey-SEM	survey construct (AI adoption)
S24	Alam et al. (2025)	International Journal of Hospitality Management (3)	Malaysia	336 hospitality-industry respondents	survey-SEM	survey construct (AI adoption + technology readiness)
S25	Alwakid & Dahri (2025)	Technology in Society (2)	Saudi Arabia	250 SMEs (managers/owners), manufacturing + services	survey-SEM	survey construct (AI capabilities: infrastructure, business integration, proactive stance)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S26	Banna & Alam (2025)	International Journal of Entrepreneurial Behaviour and Research (3)	EU	1,479 firms in 26 EU countries, 2012-2023	panel econometrics	AI venture capital invested in the firm's country (same value for every firm in that country)
S27	Basnet et al. (2025)	International Review of Financial Analysis (3)	USA	US firms, annual reports (10-K) 2005–2018	panel econometrics	annual-report text (AI mentions classified as actionable, speculative, or irrelevant)
S28	Bin-Nashwan & Li (2025)	Technology in Society (2)	China	433 Chinese accounting professionals, survey 2023–2024	survey-SEM	survey construct (AI-infused knowledge systems)
S29	Cao et al. (2025)	European Management Review (3)	USA	140 US retail managers; instrument built from 6,519 AI documents of 37 retailers	fsQCA	survey instrument (AI integration at task, function, and firm level)
S30	Chakraborty (2025)	Current Issues in Tourism (2)	India	408 hotel and resort managers, India, two-wave survey	survey-SEM	survey construct (GenAI adoption)
S31	Chiu et al. (2025)	Business Ethics the Environment and Responsibility (2)	Taiwan	33 listed semiconductor firms	DEA	patents (AI-driven innovation proxied by patent activity)
S32	D'Amico et al. (2025)	Journal of Technology Transfer (3)	UK	14,143 UK firms, 2004-2020	panel econometrics	archival records (AI technology or product use identified from a commercial database)
S33	Feng et al. (2025)	Quarterly Review of Economics and Finance (2)	China	4,004 Chinese listed firms, 2011–2023	panel econometrics	procurement index (AI adoption from procurement records)
S34	Fontanelli et al. (2025)	Journal of Economic Behavior and Organization (3)	France	French firms, 2019 survey on information and communication technology use	panel econometrics	survey (predictive AI use; buyers vs in-house developers)
S35	Guo et al. (2025)	International Marketing Review (3)	China	322 SMEs using cross-border platforms	survey-SEM	survey construct (AI adoption coupled with strategic agility)
S36	Huang & Lin (2025)	Managerial and Decision Economics (2)	USA	50 S&P 500 AI adopters plus non-adopter controls, 2010–2017	panel econometrics	news announcements (manually verified AI implementation)
S37	Kumar et al. (2025)	Journal of Business Research (3)	multi-country	277 B2B managers, multi-country survey	survey-SEM	survey construct (GenAI adoption)
S38	Mehta et al. (2025)	Journal of Business and Industrial Marketing (2)	India	496 survey respondents plus 26 key-account-manager interviews, automobile sector	mixed	survey construct (adoption of AI technologies by key account managers)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S39	N. & Roy (2025)	European Journal of Marketing (3)	Australia	335 companies across industries	survey-SEM	survey construct (AI-based stakeholder engagement)
S40	Sandeep et al. (2025)	Journal of Intellectual Capital (2)	India	290 HR professionals in talent acquisition, South India	survey-SEM	survey construct (AI adoption in recruitment)
S41	Shi et al. (2025)	International Review of Financial Analysis (3)	China	Chinese listed companies, 2010–2022 (~22,900 firm-years)	panel econometrics	annual-report text (dictionary of AI-related terms)
S42	Song et al. (2025)	Industrial Management and Data Systems (2)	USA	120 AI service-failure events at US-listed companies, 2010–2024	event study	announcements (AI service-failure events)
S43	Tao et al. (2025)	Journal of Business and Industrial Marketing (2)	China	291 B2B manufacturing firms, multi-wave, multi-source survey	survey-SEM	survey construct (big-data analytics assimilation and AI capabilities)
S44	Tingbani et al. (2025)	International Journal of Finance and Economics (3)	USA	1,950 US firms, 1996–2016	panel econometrics	AI hiring intensity: share of AI skills in the firm's workforce, from job postings and employee-profile (resume) data
S45	Adams et al. (2026)	Journal of Monetary Economics (4)	USA	US firms matched to job postings, 2010–2024	panel econometrics	job postings (openings combining AI skills and pricing work signal AI-pricing adoption)
S46	Agag et al. (2026)	Tourism Management (4)	UK	1,206 UK listed tourism, travel, and hospitality firms, 2018–2024	panel econometrics	AI human capital (AI-skilled workforce measure)
S47	Alnofeli et al. (2026)	International Journal of Information Management (2)	Australia	205 banking employees	survey-SEM	survey construct (AI-powered CRM capability)
S48	Arshad et al. (2026)	International Journal of Information Management (2)	multi-country	two cross-sectional surveys of large firms worldwide	survey-based regression (two cross-sectional surveys)	survey construct (joint implementation of robotic process automation and AI)
S49	Bai et al. (2026)	Pacific Basin Finance Journal (2)	China	Chinese listed firms; AI pilot-zone policy as natural experiment	panel econometrics	policy shock (location in a government AI pilot zone as driver of AI investment)
S50	Chen et al. (2026)	Journal of Empirical Finance (3)	USA	US firms	panel econometrics	earnings-call text (share of AI terminology in management remarks)
S51	Dinh (2026)	Industrial Marketing Management (3)	Vietnam	261 manufacturing SMEs	mixed	survey construct (AI-enabled knowledge capabilities)
S52	Filieri et al. (2026)	Journal of Product Innovation Management (4)	China	Chinese listed manufacturing exporters, 2015–2022	panel econometrics	archival AI integration measure (adoption induced by tariff exposure)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S53	Giordino et al. (2026)	Technological Forecasting and Social Change (3)	Europe	432 European listed firms, 2015-2023 (financial firms excluded)	panel econometrics	archival score of organizational AI focus (commercial database rating)
S54	Kazakis (2026)	Scottish Journal of Political Economy (2)	USA	US listed firms	panel econometrics	labor-based measure (AI investment inferred from the firm's AI-related workforce)
S55	Li et al. (2026)	International Journal of Production Economics (3)	China	218 firms in supply chains	survey-SEM	survey construct (responsible AI adoption)
S56	Li et al. (2026)	Journal of Business Research (3)	China	223 firms, cross-industry	survey-SEM	survey construct (GenAI affordances: organizational memory, collaboration, creation)
S57	Lin et al. (2026)	Emerging Markets Finance and Trade (2)	China	Chinese listed firms, 2007-2022	panel econometrics	annual-report text (AI adoption from AI-related wording)
S58	Lin et al. (2026)	International Review of Financial Analysis (3)	China	Chinese listed firms	panel econometrics	annual-report text (AI keyword dictionary)
S59	Liu et al. (2026)	International Review of Financial Analysis (3)	China	Chinese publicly listed firms, 2007-2024	panel econometrics	annual-report text (frequency of AI-related keywords)
S60	Liu et al. (2026)	Technology in Society (2)	Asia-Pacific	420 firms (survey) plus a vignette experiment with 360 managers	mixed	survey construct (AI-enabled innovation); experimentally manipulated regulatory conditions
S61	Mukherjee et al. (2026)	International Journal of Quality and Reliability Management (2)	India	312 responses from luxury hotels	survey-SEM	survey construct (AI adoption intention)
S62	Patel (2026)	Technological Forecasting and Social Change (3)	USA	1,306 AI-scored service patents from 3,899 manufacturing firms, 1980-2020	event study	patents (AI intensity of service-technology patents)
S63	Renfei et al. (2026)	Technovation (3)	China	292 responses from 120 manufacturing firms	survey-SEM	survey construct (perceptive, predictive and prescriptive AI capabilities)
S64	Singh et al. (2026)	Business Strategy and the Environment (3)	USA	S&P 500 firms, 2000-2024	panel econometrics	patents (AI patents granted per year)
S65	Song et al. (2026)	Finance Research Letters (2)	USA	US firms, 2018-2023	panel econometrics	disclosure-vs-patents gap (AI claims not backed by AI patents, i.e. AI washing)
S66	Ullah et al. (2026)	International Journal of Information Management (2)	China	419 responses from 290 manufacturing firms	survey-SEM	survey construct (AI adoption)
S67	Wang & Zhang (2026)	International Journal of Information Management (2)	China + Europe	357 early-stage ventures, two-wave survey	mixed	survey construct (AI innovation capability)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
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Table A.2: Included studies: outcomes, effect directions, and conditions (n = 67). Source: author's compilation.

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S01	Wamba-Taguimdje et al. (2020)	Perf.	Perf: organizational and process performance (qualitative)	conditional	process reconfiguration required; complementary resources (data, talent, domain knowledge, partnerships, infrastructure)	AI transformation projects improve organizational and process performance, but only when firms use AI to reconfigure their processes rather than overlay it on existing ones.
S02	Baabdullah et al. (2021)	Perf.	Perf: perceived SME performance (survey scale)	positive	adoption enablers: technology roadmapping, attitude, infrastructure readiness, awareness; professional expertise not significant	Acceptance of AI practices improves perceived SME performance; adoption is driven by roadmapping, attitude, infrastructure and awareness - not by professional expertise.
S03	Chatterjee et al. (2021)	Both	Perf: perceived organizational performance; CA: competitive advantage scale (incl. market-share gain)	positive	mediators: B2B engagement, employee experience, information processing; moderator: leadership support	AI-CRM implementation improves perceived organizational performance and competitive advantage in B2B relationship management.
S04	Chatterjee et al. (2022)	Perf.	Perf: firm performance scale; B2B relationship satisfaction	positive	technology turbulence weakens the paths to relationship satisfaction; leadership support strengthens the satisfaction-to-performance path	AI-embedded CRM improves relationship satisfaction and firm performance; gains shrink under technology turbulence and grow with leadership support.
S05	Hossain et al. (2022)	CA	CA: sustained competitive advantage (relative performance vs. competitors)	conditional	mediators: market sensing, seizing, and reconfiguring carry the effect in full; moderator: AI adoption strengthens all three capability paths	Marketing analytics capability drives sensing/seizing/reconfiguring toward sustained competitive advantage; AI adoption amplifies this only when built on an analytics foundation.
S06	Lee et al. (2022)	Perf.	Perf: revenue growth	conditional	adoption-intensity threshold (low-level adoption shows no effect); complementary cloud and database investments amplify	AI pays off only above an adoption-intensity threshold, and more so with complementary technology investments and venture-specific internal R&D.
S07	Leoni et al. (2022)	Perf.	Perf: perceived manufacturing firm performance	conditional	full mediation: knowledge management processes carry the entire effect, the direct path from AI to performance is not significant; AI has no direct effect on supply chain resilience either; larger firms reach higher AI maturity	AI alone does not improve manufacturing firm performance - the effect exists exclusively through knowledge management processes (full mediation).

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S08	Lui et al. (2022)	Perf.	Perf: firm market value (abnormal stock returns around the announcement)	negative	stronger negative reaction for nonmanufacturing firms, weak IT capability and low credit ratings; better IT capability lessens the drop	Investors react negatively to AI investment announcements on average; the penalty concentrates in nonmanufacturing firms with weak IT capability and low credit ratings.
S09	Mishra et al. (2022)	Perf.	Perf: gross and net operating efficiency, net profitability, advertising spending, employment	mixed		AI focus reduces gross operating efficiency but increases net operating efficiency and profitability - automation raises costs more slowly than revenues.
S10	Sun et al. (2022)	CA	CA: perceived competitive advantage and innovation intelligibility	conditional	full mediation: value co-creation and the economic and relationship value it creates have no direct effect; dynamic and innovation capabilities carry the entire effect	Value co-creation in the AI innovation ecosystem has no significant direct effect on competitive advantage - the effect runs through dynamic and innovation capabilities (full mediation).
S11	Ali et al. (2023)	Both	Perf: clinical, financial, organizational and technological outcomes (qualitative, financial gains quantified); CA: competitive position vs. rivals (patient choice, payer bargaining power, differentiation)	positive	prerequisites: qualified robotic and revenue-cycle team, cultural shift in the medical community, regulatory approval; barrier: high acquisition costs not covered by insurance	AI-enabled robotic surgery improves clinical, financial and technological outcomes and strengthens the competitive position of the healthcare organization.
S12	Czarnitzki et al. (2023)	Perf.	Perf: firm productivity (sales- and value-added-based, production function)	positive		AI adoption is robustly associated with higher firm productivity in representative German data; the effect grows with usage intensity.
S13	Huang & Lee (2023)	Perf.	Perf: firm market value (stock-price reaction around the announcement)	positive	detailed announcements pay more overall but react with delay - vague ones only on the day itself; IT firms gain more, non-IT reactions not significant; negative wording correlates modestly negatively	Markets reward AI implementation announcements - more so when announcements are detailed (delayed but stronger reaction) and come from IT firms; non-IT reactions not significant.
S14	Krakowski et al. (2023)	Both	Perf: competitive performance (game outcomes/ratings); CA: persistent performance differences (sustained advantage) across settings	conditional	substitution: traditional human capabilities become obsolete; complementation: new human-machine capabilities emerge that are unrelated or negatively related to traditional ones	AI adoption shifts the sources of competitive advantage: a new decision-making resource emerges at the human-AI intersection, unrelated to traditional capability.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S15	Samadhiya et al. (2023)	Both	Perf: perceived firm performance (survey scale, expectation-worded items); CA: perceived sustainable competitiveness (survey scale; only one of five items competitor-referencing, rest mixes AIPRM embedding and environmental commitment)	positive	prerequisites: information and communication technology capability and technological readiness; firm fit	AI-based partner relationship management improves firm performance and sustainable competitiveness, provided information and communication technology capability and technological readiness are in place.
S16	Wu & Monfort (2023)	Perf.	Perf: perceived firm performance (survey scale)	positive	configurational: marketing capabilities, customer value co-creation, market orientation and AI strategy jointly form necessary and sufficient recipes for high performance	AI marketing strategy raises firm performance, but only as part of configurations with marketing capabilities, co-creation and market orientation - a configurational result.
S17	Babina et al. (2024)	Perf.	Perf: growth in sales, employment, and market valuation	positive	channel: product innovation rather than process efficiency; gains concentrate among firms that were already large, which raises industry concentration (superstar dynamics)	AI-investing firms grow in sales, employment and value via product innovation; benefits concentrate among large firms and go along with rising industry concentration.
S18	Cannas et al. (2024)	Perf.	Perf: case-reported competitiveness outcomes: costs, lead times, service level, quality, safety, sustainability (partly quantified: 30% stock reduction, 4% OEE gain)	positive	barriers as boundary conditions: data quality, AI skills, high investment needs, unclear economic benefits, lack of AI cost-analysis experience (9 barriers total in Table 6: financial, organisational, strategic, technological)	AI improves supply-chain competitiveness, mostly evidenced in production processes; the gains depend on overcoming data-quality, skill and investment-evaluation barriers.
S19	Pesqueira et al. (2024)	Both	Perf: operational efficiency in tender management (processing times, decision quality, document accuracy), expert-rated; CA: expert-rated comparative competitiveness of the four companies in the tendering context	conditional	enablers: governance and ethical frameworks, regulatory compliance, organizational readiness, alignment with business goals; senior-management involvement identified as critical	AI adopters manage pharmaceutical tendering faster and with better decisions than non-adopters, but only sustainably within robust governance and compliance structures.
S20	Pham et al. (2024)	Perf.	Perf: outpatient revenue, inpatient revenue, ROA, productivity, occupancy (ROA not significant in every specification)	mixed	larger-market-share hospitals are the adopters (self-selection, no test of who benefits more); without accounting for it the estimates mislead	Hospitals with larger market share adopt AI and gain in revenue, productivity and occupancy; naive comparisons that ignore which hospitals adopt AI misestimate the effects.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S21	Sullivan & Fosso (2024)	Perf.	Perf: perceived firm performance, process innovation, product innovation	conditional	central mediator: adaptive response to market changes; only AI analytics shows direct effects in a supplementary check; hostility weakens the automation path, strengthens the analytics path; only ten of eighteen conditional effects significant	AI capabilities pay off through the firm's adaptive response to market changes; environmental hostility/dynamism condition how much each AI capability contributes.
S22	Sun et al. (2024)	Perf.	Perf: firm total factor productivity	positive	stronger for state-controlled, internationally oriented and innovative firms; channels: reduced information asymmetry (key), specialized division of labor and green innovation only potential; supply-chain digitalization policy amplifies; the effect weakens over time	AI raises firm productivity in China - most for state-controlled, international and innovative firms - but the productivity boost decays over time.
S23	Zebec & Indihar (2024)	Perf.	Perf: organizational performance (survey scale)	positive	mediators in sequence: decision-making quality and business-process performance carry the effect; process automation, organizational learning and process innovation as path-specific partial mediators	AI creates business value indirectly: through better decisions and business-process performance, complemented by automation, learning and process innovation.
S24	Alam et al. (2025)	Both	Perf: value creation (survey scale); CA: sustainable competitive advantage (modeled as mediator)	positive	mediator: sustainable competitive advantage between AI adoption and value creation; dynamic capabilities have a negative direct effect on value creation (transition costs); technological turbulence moderates only the capabilities-to-value path, weakening it further	AI adoption creates value in hospitality directly and via sustainable competitive advantage; dynamic capabilities show a negative direct effect on value creation (transition costs), aggravated under technological turbulence.
S25	Alwakid & Dahri (2025)	Perf.	Perf: sustainable SME performance (survey scale)	positive	mediators: SME creativity and green innovation; green entrepreneurial orientation as parallel driver	AI capabilities raise sustainable SME performance through creativity and green innovation; AI infrastructure, proactive strategy and entrepreneurial risk-taking are the key resource inputs.
S26	Banna & Alam (2025)	Perf.	Perf: firm revenue growth	conditional	U-shape: AI hurts revenue below and helps above roughly USD 11.3M; works only when paired with own R&D; only small and medium firms benefit	AI investment follows a U-shaped payoff: short-term revenue drag, long-term gains - and only firms coupling AI with R&D strategy capture the upside.

Table A.2 (continued)

ID	Study			Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S27	Basnet	et al.	Perf.		Perf: firm value (market valuation); R&D, patents, and later productivity as channels	conditional	disclosure substance decides: actionable AI disclosures bring valuation gains, innovation, and later productivity; speculative or irrelevant ones bring nothing; silent or vague peers are penalized	Markets reward only substantive (actionable) AI disclosures - early adopters with real implementation plans gain value via innovation; vague AI talk earns nothing.
S28	Bin-Nashwan & Li		Perf.		Perf: perceived accounting-firm performance	conditional	mediators: green human capital and green structural capital carry the effect (no direct path modeled); green relational capital channel empty; moderator: sustainability culture	AI-infused knowledge improves accounting-firm performance through green human and structural capital, amplified by sustainability culture; relational capital plays no role.
S29	Cao et al.	(2025)	Perf.		Perf: efficiency and innovation outcomes (configurational)	conditional	configurational: no single AI function suffices; performance needs combinations, especially customer service and cybersecurity paired with other functions; emergent capabilities: adaptive learning, predictive analytics, uncertainty mitigation	Retail performance comes from combinations of AI-enabled functions, not from isolated AI uses; a directly configurational result.
S30	Chakraborty		Both		Perf: firm/organisational performance (survey scales); CA: competitive advantage / competitive positioning (survey scale)	positive	enabler: technological readiness; moderator: leader narcissism shapes the adoption drivers and the effect of generative AI on competitive positioning	GenAI adoption improves hotels competitive advantage and performance, with leadership personality (narcissism) shaping how adoption translates into outcomes.
S31	Chiu et al.	(2025)	Perf.		Perf: profit, sustainability, and market-valuation efficiency	mixed	stage-specific: AI innovation lifts sustainability and market efficiency but not profit efficiency (size and leverage drive profit); firm scale raises profit efficiency yet lowers sustainability and market efficiency	AI-driven innovation lifts sustainability and market-valuation efficiency in semiconductors but not short-term profit efficiency - the payoff is stage-specific.
S32	D'Amico	et al.	Perf.		Perf: innovation performance (knowledge spillover of innovation)	conditional	complement: own R&D investment (AI pays off only when paired with internal R&D); AI substitutes for knowledge collaboration with certain external partners	AI adoption boosts innovation only when paired with the firms own R&D investment - and partly replaces external knowledge collaboration.
S33	Feng et al.	(2025)	Perf.		Perf: innovation output and innovation efficiency	positive	stronger for large incumbents and growth-stage firms; executives with IT backgrounds amplify; highly innovative firms diversify supply chains to reduce resource risk	AI adoption raises innovation output and efficiency, most where firm maturity, scale and IT-savvy leadership provide absorptive conditions.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S34	Fontanelli et al. (2025)	Perf.	Perf: volatility of productivity growth (not its level)	conditional	buyers of external AI see higher productivity volatility, in-house developers none; complementary human capital (share of technology engineers and technicians) softens the buyer effect	Off-the-shelf AI raises the riskiness of productivity outcomes; developing in-house or employing complementary technical staff neutralizes it.
S35	Guo et al. (2025)	CA	CA: international advantage: product innovation and brand upgrading (global value-chain position)	positive	mediator: international marketing capabilities; platform embeddedness shows inverted-U effects (depth on product innovation, breadth on brand upgrading); medium embeddedness is optimal	AI adoption coupled with strategic agility upgrades SMEs global value-chain position through marketing capabilities - but only at moderate platform embeddedness (inverted-U).
S36	Huang & Lin (2025)	Perf.	Perf: financial ratios, productivity, market value (Tobin's q)	mixed	indicator split: financial performance and market value improve, productivity does not; first movers show partly negative or insignificant effects (no uniform first-mover advantage)	AI implementation lifts financial performance and market value, but first movers and top performers do not benefit uniformly across all indicators.
S37	Kumar et al. (2025)	Perf.	Perf: perceived firm performance (survey scale)	positive	moderator: ethical leadership on the adoption-to-performance path; adoption drivers and barriers: uniqueness, information completeness, convenience, deceptiveness	GenAI adoption boosts perceived firm performance, more strongly under ethical leadership.
S38	Mehta et al. (2025)	Both	Perf: firm performance (survey scale); CA: competitive advantage (modeled as mediator)	positive	manager personality traits (Big Five) drive adoption; competitive advantage mediates between adoption and performance; organizational culture moderates the effect of agreeableness on adoption	Individual personality traits shape AI adoption in key account management, which converts into firm performance through competitive advantage - culture conditions the path.
S39	N. & Roy (2025)	Perf.	Perf: firm performance (survey scale)	positive	mediator: marketing agility; moderator: technological turbulence amplifies the positive effects	AI-based stakeholder engagement improves performance through marketing agility - and the payoff is largest in technologically turbulent environments.
S40	Sandeep et al. (2025)	CA	CA: perceived competitive advantage from AI-enabled recruitment (survey scale)	positive	HR competencies and open innovation build dynamic capabilities, which mediate; financial support and IT infrastructure act as enabling resources	AI adoption in recruitment yields competitive advantage only where dynamic capabilities, built from HR competencies and open innovation with financial/IT support, are in place.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S41	Shi et al. (2025)	Perf.	Perf: enterprise value (Tobin's q)	positive	data assets mediate 45.6% of the effect while a significant direct effect remains; managerial capability moderates; boundary conditions: no effect in heavily polluting industries unless the firm has a credible green transition, and none in asset-intensive firms; state-owned enterprises gain about twice as much as private ones	AI adoption raises enterprise value through data assets, and only under capable management and agile (non-asset-heavy) structures - the study explicitly maps when AI becomes a source of advantage.
S42	Song et al. (2025)	Perf.	Perf: firm value (stock-price reaction)	negative	small firms hit harder; losses more negative in recent years; damage ranked by failure type: accuracy worst, then safety, privacy, fairness	AI service failures destroy market value - most for small firms and accuracy failures; the downside risk of AI implementation is real and priced.
S43	Tao et al. (2025)	Perf.	Perf: new product performance (survey scale)	positive	mediator: AI capabilities (big-data analytics assimilation alone is insufficient); moderator: electronic supply-chain collaboration strengthens the indirect effect	Big-data/AI assimilation lifts new product performance only when converted into AI capabilities, and more so with electronic supply-chain collaboration.
S44	Tingbani et al. (2025)	Perf.	Perf: firm growth (sales growth)	positive	labour productivity amplifies the growth effect; labour cost and labour share weaken it	AI investment raises firm growth modestly on average; labour-market conditions moderate the size of the gain - productivity-rich, labour-cost-light firms capture more of it.
S45	Adams et al. (2026)	Perf.	Perf: growth in sales, employment, assets, markups; stock-return response to monetary policy	positive	adoption concentrated in larger, more productive firms; adopters more responsive to monetary policy surprises	Firms adopting AI-powered pricing grow faster in sales, employment and markups - AI pricing is a capability large productive firms exploit first.
S46	Agag et al. (2026)	Perf.	Perf: firm value; wage inequality as mediating channel	positive	wage inequality partially mediates: AI human capital compresses inequality and thereby raises firm value; the effect is stronger in dynamic and complex industries; by sector, the value effect is strongest in travel and the inequality compression in hospitality	AI human capital raises firm value partly by compressing wage inequality - a workforce-centered pathway to AI-enabled advantage, strongest in dynamic sectors.
S47	Alnofeli et al. (2026)	Both	Perf: profitability (survey scale); CA: competitive advantage (survey scale)	positive	mediator: marketing ambidexterity (balancing exploration and exploitation) carries the effect	AI-CRM capability lifts profitability and competitive advantage through marketing ambidexterity - capability without ambidexterity closes the value-realisation gap only partially.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S48	Arshad et al. (2026)	Perf.	Perf: revenue growth; cost reduction	mixed	combining robotic process automation and AI adds revenue growth via price increases but no extra cost reduction (each technology alone does cut costs); a strategic revenue-growth intent amplifies the revenue gains	Combining robotic process automation and AI pays through higher revenue, not lower cost — and only for firms whose strategic intent targets growth.
S49	Bai et al. (2026)	Perf.	Perf: stock price crash risk; firm value	mixed	crash risk rises through lower information transparency and managerial over-optimism (mechanisms); worse for low-transparency and resource-constrained firms; value gains may reflect short-term market optimism	Policy-driven AI investment inflates firm value but simultaneously raises crash risk - short-term valuation gains trade off against long-term stability.
S50	Chen et al. (2026)	Perf.	Perf: investment efficiency; Tobin's q (long horizon)	positive	mechanisms: better sales forecasts, reporting quality, and product/process innovation; exposure to data-privacy regulation weakens the effect	AI adoption improves how firms allocate capital - unless regulatory friction (data-privacy exposure) blunts it; value shows up in Tobins q only over longer horizons.
S51	Dinh (2026)	Perf.	Perf: new product performance (survey scale)	positive	direct effect plus a path through more radical digital product innovation (mediator); digital sustainability leadership acts as both mediator and moderator	AI knowledge capabilities lift new product performance in emerging-market SMEs, with digital sustainability leadership as the critical boundary condition.
S52	Filieri et al. (2026)	Perf.	Perf: supply chain efficiency, sales performance, market value	conditional	inverted U-shape: AI benefits diminish at extreme tariff exposure (an optimal investment threshold); customer engagement needs strengthen the effect; tariff exposure itself induces AI integration	Trade tensions push firms into AI integration that improves efficiency, sales and value - up to a threshold where adversity overwhelms the compensation.
S53	Giordino et al. (2026)	Perf.	Perf: ROA; Tobin's q; ESG pillar scores	positive		Organizational AI focus goes hand in hand with better financial performance and better environmental and social sustainability scores in European listed firms.
S54	Kazakis (2026)	Perf.	Perf: firm efficiency	conditional	no unconditional effect; gains only with capable managers, stronger competitive pressure, stable institutional ownership, and access to long-term debt	AI investment alone does nothing for efficiency - gains materialize exclusively under managerial, competitive, ownership and financing conditions ("Conditional Gains").

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S55	Li et al. (2026)	CA	CA: perceived competitive advantage (survey scale)	conditional	full mediation: no significant direct path; the effect runs entirely through distributive and procedural justice (fairness of outcomes and of process); supply chain complexity weakens the distributive channel but not the procedural one	Responsible AI builds competitive advantage through supply-chain justice - in complex networks, only procedural fairness remains a robust channel.
S56	Li et al. (2026)	CA	CA: perceived sustained competitive advantage (survey scale)	positive	three GenAI affordances mediate between technological opportunism (a firm's readiness to seize new technology) and competitive advantage; competitive intensity amplifies only the creation-related path	A firm's readiness to seize new technology converts into competitive advantage through what GenAI enables; under intense competition, only the creative use of GenAI keeps paying.
S57	Lin et al. (2026)	Perf.	Perf: total factor productivity	positive	internal control quality partially mediates: AI improves governance and risk management, which in turn raises total factor productivity	AI raises firm productivity partly by improving internal control quality - governance is a transmission channel, not just a moderator.
S58	Lin et al. (2026)	Perf.	Perf: investment efficiency; firm value	positive	stronger in knowledge-intensive and R&D-intensive firms and firms with sound internal controls; mechanisms: innovation capability and financial health	AI enhances investment efficiency and firm value most where knowledge intensity, R&D and internal controls give the firm the capacity to absorb it.
S59	Liu et al. (2026)	Perf.	Perf: likelihood of M&A; long-term firm value	positive	stronger with abundant cash holdings, less-developed market environments, private ownership	AI adoption makes firms more acquisitive and raises long-term value - internal resources and institutional context set the boundary.
S60	Liu et al. (2026)	Perf.	Perf: productivity gains + firm growth	positive	regulatory clarity and supportive enforcement strengthen the payoff; confirmed causally in an experiment	AI-enabled innovation converts into productivity and growth only under clear, supportive regulation - institutional environment is the amplifier.
S61	Mukherjee et al. (2026)	Perf.	Perf: service performance (survey scale)	positive		AI adoption intention lifts hotel service performance; drivers: ease of use, competitive pressure, policy support - the infrastructure estimate is negative.
S62	Patel (2026)	Perf.	Perf: stock market reaction to patent grants	negative	firm-specific stock volatility softens the negative reaction (moderator); asset tangibility does not; the effect appears only in the 2010s and 2020s	Markets react negatively to AI-heavy service patents in manufacturing — a caution against assuming AI-based service moves create short-term market value.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S63	Renfei et al. (2026)	Perf.	Perf: corporate sustainability performance including an economic dimension (cost reduction, revenue from waste, spin-offs)	positive	prescriptive (action-recommending) capabilities fail as part of the AI capability bundle; firm type is a boundary condition: strong effect for Industry 4.0 (digitally advanced) manufacturers, much weaker but not absent for traditional firms	AI capabilities raise sustainability and economic performance mainly in digitally advanced (Industry 4.0) manufacturers; action-recommending (prescriptive) AI fails, and traditional firms barely benefit.
S64	Singh et al. (2026)	Perf.	Perf: ROA; firm value	conditional	small positive ROA effect, stronger for R&D-intensive firms; firm-value gains only for firms with good sustainability records; stronger under Democratic administrations (political environment as condition)	AI innovation pays modestly and unevenly: R&D intensity, sustainability credentials and even the political climate condition whether markets reward it.
S65	Song et al. (2026)	Perf.	Perf: market reactions; long-term operating performance	negative	time path: short-lived positive investor reaction, then negative long-run stock returns and sustained underperformance; the gap between claims and patents is what drives the effect	AI talk without AI substance backfires: markets initially reward the narrative, then punish rhetoric that outpaces implementation.
S66	Ullah et al. (2026)	Perf.	Perf: organizational performance (survey scale)	positive	team dynamics, corporate innovation capabilities, high-commitment workforce (mediators)	AI adoption converts into performance where cohesive teams, innovation capability and a committed workforce absorb it - human/organizational enablers unlock the value.
S67	Wang & Zhang (2026)	Perf.	Perf: entrepreneurial success (survey scale)	conditional	full mediation: strategic resilience fully carries the effect of AI capability on venture success; configurational analysis finds several equally effective condition combinations for high performance	AI capability makes ventures successful only via organizational resilience - the mechanism is organizational, not technological; multiple configurations lead there.

List of Aids

The following AI tools and aids were used during the preparation of this thesis.

Global (all sections)

Section	Tool	Description
ALL	Grammarly Pro	For mistake and style checking throughout the thesis. Mistakes were corrected immediately; style suggestions were reviewed manually and applied when suitable.
ALL	Claude Code (Anthropic)	Used throughout for LaTeX text correction and conversion, literature management (renaming PDFs, managing BibTeX), file structure, and drafting/editing section content. All AI-generated text was reviewed and approved before inclusion.

Section	Tool	Description
ALL (writing phase)	library-rag — local retrieval tool over the thesis corpus (built with Claude Code, 27 July 2026)	Local search system over the 67 full texts of the frozen corpus, used from the writing phase onward to locate citable passages. Runs entirely on the author’s machine, no cloud services: Docling for PDF parsing, Qwen/Qwen3-Embedding-4B (fp8) for semantic search, BAAI/bge-reranker-v2-m3 for reranking, hybrid semantic and BM25 retrieval fused by reciprocal rank fusion. The tool retrieves and locates passages; it does not generate thesis text. Every verbatim quote it returns passes a two-stage programmatic check before it may enter a footnote: the quote must appear verbatim in the retrieved source passage, and its words must appear in the same order on the cited page of the PDF itself. The second stage also catches a quote attributed to the wrong passage or page. Its threshold was calibrated against 40 genuine and 60 fabricated quotes from this corpus (genuine: median 1.00; fabricated: maximum 0.18; the threshold of 0.60 admits none of the fabricated ones). Quotes the check cannot confirm are flagged for manual inspection of the page rather than passed through. Printed page numbers are calibrated per document from the printed page labels in the PDF and cross-checked against the Scopus page range; where no printed page number exists, the tool states this instead of supplying one. Corpus identification, screening, and coding were completed and frozen (17–18 July 2026) before this tool existed and are unaffected by it.

Section	Tool	Description
ALL (writing phase, from 13 Aug 2026)	Antigravity CLI (Google)	Successor backend for the blind citation checks: Google discontinued the Gemini CLI sign-in for individual accounts (it still worked on 28 July 2026 and was refused by 13 August 2026), so <code>corpus/gemini_verify_citations.py</code> now calls the Antigravity CLI (model: Gemini 3.1 Pro) instead of Gemini 2.5 Pro. The blind design is unchanged — the model still receives nothing but the sentence and the footnote passage, and the tool now additionally runs in an empty working directory so it cannot read any project file. Each verdict record notes the model that produced it; all verdicts from before the switch remain Gemini 2.5 Pro and are labelled as such. The migrated script was verified with a deliberately overclaiming test citation, which the new model flagged correctly.
Proposal	Claude Code (Anthropic)	Readability rewrite pass over the proposal prose; minimal simplifications merged into <code>proposal_final.tex</code> after review; citations, quotes, and footnotes left unchanged.
Proposal	Quillbot Paraphraser	Comparison paraphrase of citation-free proposal sentences (<code>proposal_quillbot.tex</code>) to evaluate phrasing alternatives; citations, quotes, and footnotes left unchanged.
Proposal	Claude Code (Anthropic)	Source check of the proposal against the collected article PDFs: every cited claim compared with the source text, page numbers and verification footnotes added to all citations, and five wordings corrected where they had drifted from the source. All corrections reviewed before merging.
Proposal	Claude Code (Anthropic)	Shortening pass after I found the draft too dense: text condensed to 3 pages, one main statement per source, detail sentences removed. Citations and quotes kept verbatim; result reviewed by me.

Section	Tool	Description
Method	Claude Code (Anthropic)	Corpus retrieval (17 July 2026): wrote and ran the script <code>corpus/pull_corpus.py</code> , which executes the documented Scopus query via the Scopus API (WU license) and saves the frozen export (432 records, raw JSON + CSV). Verified that all three benchmark studies are retrieved and that record counts match Scopus.
Method	Claude Code (Anthropic)	Title/abstract pre-screening (17 July 2026): Claude read the full abstracts of all 174 studies passing the $AJG \geq 2$ filter and proposed include/exclude decisions with reason codes against the criteria catalogue approved beforehand; borderline cases flagged for full-text checking. All proposals recorded in <code>corpus/screening_2026-07-17.csv</code> and reviewed by the author before becoming final.
Method	Claude Code (Anthropic)	Full-text screening assistance (17–18 July 2026): Claude extracted and summarized method/measurement sections from the PDFs of all flagged borderline studies and proposed verdicts against the criteria catalogue; 12 studies excluded content-based, 1 confirmed as include; two inaccessible reports excluded as not retrieved (author’s decision). Claude also matched, renamed, and de-duplicated the downloaded PDFs. All verdicts recorded with reasons and reviewed by the author.

Section	Tool			Description
Results (coding)	(coding)	Claude (Anthropic) Gemini (Google)	Fable 5 + CLI	Dual independent AI coding of all 67 included studies against the frozen coding scheme (18 July 2026): Claude Fable 5 drafted all rows from the full texts (coder 1); Gemini 2.5 Pro coded the same studies blind—receiving only the coding scheme and PDF text, never Claude’s drafts (coder 2). A documented self-consistency pass harmonized coder 1’s effect-direction codes before comparison. Inter-coder agreement: method 90%, outcome construct 88%, effect direction 73%.
Results (coding)	(coding)	Claude (Anthropic)	Fable 5	Adjudication support (18 July 2026): for all 29 coder disagreements and 10 random-sample agreement rows, Claude Fable 5 performed full-text reads and prepared evidence briefs quoting both coders’ positions with page references; 15 factual gaps resolved from the full texts. The author checked the coded values of all 67 studies against the results reported in the papers, including the rows on which both runs agreed, and read the full text of each of the 29 disagreements to reach an independent decision before consulting either run’s values; the cases were then resolved against the briefs and the written decision rules, with decisions recorded per case. No joint coder errors found; all 67 rows finalized after the author’s batch approval.

Section	Tool		Description
Theory (Section 2.1)	Claude (Anthropic) library-rag	Code +	Drafting of Section 2.1 on the construct distinction (28 July 2026): citable passages located with the local retrieval tool; all four citations checked two-stage (verbatim in the source passage, word sequence on the cited PDF page) and their printed page numbers confirmed (pp. 215, 247, 550). Extended on 28 July 2026 by Barney’s (1991, p. 102) definitions of competitive advantage and sustained competitive advantage, taken from the verified text sidecar and read off the page image; the corpus retrieval tool does not cover sources outside the corpus. The counts come from the fact sheet derived from the coding table, not from the tool. BibTeX entries generated from the frozen Scopus export with corpus/make_bib.py. Human-voice pass applied. Text reviewed and edited by the author before inclusion.
Results (descriptive mapping)	(de-map- Claude Code (Anthropic)		Descriptive figures (18 July 2026): wrote and ran the script corpus/make_figures.py, which derives five corpus-mapping charts (publication year, method, sample geography, outcome construct $\times$ effect direction, AJG rating) directly from the finalized coding table and writes them to figures/ as PDF. Category groupings are documented in the script; all counts verified to sum to $n = 67$. Colors validated as colorblind-safe. Figures reviewed by the author before inclusion.

Section	Tool	Description
Theory (Sections 2.2–2.5)	Claude (Anthropic) library-rag	Code + Drafting of the four theory sections on general-purpose technology, the resource-based view, automation/augmentation and situated AI (28 July 2026). Citable passages from the corpus located with the local retrieval tool; all 11 corpus quotes checked two-stage (verbatim in the source passage, word sequence on the cited PDF page). Two quotes flagged by the check were inspected manually on the page; one revealed a page error (Krakowski et al., 1444 → 1445), which was corrected. The three theory anchors outside the corpus (Kemp 2024, Raisch & Krakowski 2021, Brynjolfsson et al. 2021) are not in the retrieval index; their passages were read directly from the PDFs with pdfplumber and the printed page numbers verified against the running heads. Barney (1991) is cited from the verified text sidecar to the image scan (literature/barney1991firm.txt, OCR base checked against the scan page by page); both quoted passages were additionally read off the page images before the footnotes were set, and the original’s printed typos are reproduced verbatim. BibTeX entries generated from the frozen Scopus export with corpus/make_bib.py. Human-voice pass applied, including the paraphrase check against every footnote passage. Text reviewed and edited by the author before inclusion.

Section	Tool	Description
Theory (Sections 2.1–2.5)	Claude Code (Anthropic) + Gemini CLI (Google)	Blind citation check of the finished theory chapter (28 July 2026), continuing the dual-AI principle from the coding phase. Claude wrote the script <code>corpus/gemini_verify_citations.py</code> , which parses every citation site in a section and splits the check by what can be decided mechanically: whether a quote appears verbatim on the cited page is settled deterministically by the local retrieval tool, the source-echo rule (three or more consecutive words shared between the prose and the cited passage) is computed by the script, and only the judgement of paraphrase fidelity goes to Gemini 2.5 Pro — blind, receiving nothing but the sentence and the footnote passage. Of 25 citation sites in the chapter, the script flagged 7 for shared word sequences; the verdicts are recorded per site and collected in a report. The author decides every flag; the script proposes nothing and changes no text.

Section	Tool	Description
Method / bibliography	Claude Code (Anthropic)	Citation rule for the seven studies without printed page numbers (28 July 2026): each of the seven checked against its publisher type and the rule approved by the author implemented — article number in the bib eid field plus the article-internal page for the three Elsevier articles, section references for the four with publisher-relative or not-yet-assigned pagination. Two defects in bib.bib were found and fixed in the process: an unescaped ampersand in one title (would have aborted the LaTeX run at the first citation of it) and acronyms being lowercased by APA sentence casing, which had printed “ai”, “sme” and “b2b” in the reference list. corpus/make_bib.py now escapes ampersands and brace-protects acronyms, with self-tests; the 27 existing entries were corrected in one pass and the reference list checked in the compiled PDF. One Scopus metadata error corrected against the source PDF: the author of the European Journal of Marketing article was recorded as “N., N. A.” and is Tehrani, A. N.
Results (descriptive mapping)	Claude Code (Anthropic)	Drafting of Section 4.1 (18 July 2026): Claude drafted the descriptive-mapping prose directly from the finalized coding table (aggregate counts only, no source citations) and applied the agreed human-voice pass (banned-pattern audit, voice matching). All numbers cross-checked against the coding table; text reviewed and edited by the author before inclusion.

Section	Tool	Description
Appendix (coding table)	Claude Code (Anthropic)	Reader-facing cleanup of the appendix coding table (29 July 2026): a rendering layer in <code>corpus/make_appendix_table.py</code> strips coder shorthand from the printed table — path coefficients, significance stars, hypothesis labels, model-fit values, and coder notes removed mechanically from 26 cells; 23 cells whose internal path abbreviations could not be resolved by rule were reworded into plain sentences, stored separately in <code>corpus/appendix_overrides.tsv</code> . The coded values in <code>corpus/coding_table.csv</code> are untouched — the layer changes wording only, never a category, effect direction, or the identity of a condition. Two self-checks run at every regeneration: no shorthand token may survive into the rendered table, and every number printed must appear verbatim in the coded source cell. All 23 rewordings documented side-by-side (coded vs. appendix wording) for the author’s review.

Section	Tool	Description
Method	Claude Code (Anthropic)	Revision of the Method chapter following the author’s review of the interim draft (13 August 2026), worked through question by question with the author: the AJG filter step marked as requiring no reading (ISSN matching), the full-text stage reworded to “assessed for eligibility” with the check explained, AI assistance in screening disclosed in the chapter text, the search documented as run through the Scopus API, the coding-field list split into two explained halves, all five effect-direction values and the two coding rules written out with examples, and two example citations added for the performance/competitive-advantage distinction (Chatterjee et al. 2021, p. 209; Babina et al. 2024, p. 1) — both footnote passages read from the PDFs and the printed page number verified against the running head. Every change decided by the author.
Method, Theory, Results	Claude Code (Anthropic)	AI-pattern audit (13 August 2026) over all drafted chapters against the Wikipedia “Signs of AI writing” catalogue: four wording corrections (an emphasis adverb and a three-item example list in Method, a built antithesis in Theory, a stock idiom in Results) plus one cross-chapter consistency fix (Results now states that effect directions cover four of the five direction codes defined in Method). Citations, quotes, footnotes, and all counts unchanged; factual enumerations deliberately left as they are. Corrections listed for the author’s review.

Section	Tool		Description
Method	Claude (Anthropic)	Code +	Blind citation check of the finished Method chapter (13 August 2026), first run on the migrated backend: all 3 citation sites (Rojon et al. 2021; Chatterjee et al. 2021; Babina et al. 2024) checked mechanically for source echo (0 hits) and by Gemini 3.1 Pro for paraphrase fidelity, blind. The first run flagged 2 sites; both traced to sentence-initial citation placement, which pairs the footnote with the wrong sentence — the two example sentences were restructured to the claim-then-cite convention. The re-run flagged the Chatterjee site once more, correctly: the footnote covered only the definition half of the sentence, so the H4 passage (same page) was added and the unsupported “three-item” detail dropped. Final result: 3 of 3 sites pass both checks.
Method	Antigravity (Google)	CLI	Blind section review of the finished Method chapter (13 August 2026): Gemini 3.1 Pro received only the chapter and the fact sheet and flagged 5 findings (2 consistency, 3 style). The author decided all five: 3 fixes applied (the count of 29 adjudicated disagreements clarified as studies with at least one disagreement across the five compared fields; “unconditional average effect” reworded to “sign of the average effect” to remove a collision with the fact sheet’s conditions counts; one “not only ... but also” construction removed), 2 flags rejected as substantive prose. Decisions recorded per flag.

Section	Tool		Description
Results (synthesis)	Claude (Anthropic) Antigravity (Google)	Code + CLI	Conditions clustering for the structure of Section 4.2 (13 August 2026): Claude extracted all condition entries from the finalized coding table (63 studies) and proposed six thematic clusters with per-study assignments, flagging 8 judgment cases. Gemini 3.1 Pro classified the same 8 cases blind (cluster definitions and verbatim condition entries only, no proposals). Agreement on 4 cases; the author decided all 8: one Gemini addition adopted, three rejected with reasons, the single genuine dissent resolved for consistency with the theory chapter. Assignments frozen; cluster counts are script-derived and validated against the coding table.
Results (synthesis, Section 4.2.1)	Claude (Anthropic) Antigravity (Google)	Code + CLI	Drafting of the Section 4.2 opening and Subsection 4.2.1 on complementary investments (13 August 2026), covering the 11 studies of the frozen cluster. The local retrieval tool was unavailable in this session, so all citable passages were read directly from the source PDFs with pdfplumber; 25 quoted passages across the 11 studies were verified programmatically before entering footnotes (word sequence of every quote confirmed on the cited PDF page; printed page numbers taken from the page headers, section references for the two studies without usable pagination). Corpus counts in the prose come from the fact sheet only. The blind citation check (Gemini 3.1 Pro) ran three times during drafting: the first pass flagged 10 of 12 citation sites, leading to three precision corrections in the prose and footnote extensions with the studies' sample statements; the final pass leaves one flag, the unavoidable word-sequence match on the quantified threshold "USD 11.3 million", for the author's decision. Human-voice pass applied during drafting. Text to be reviewed by the author.

Section		Tool		Description
Results thesis, 4.2.2)	(syn- Section	Claude (Anthropic) Antigravity (Google)	Code + CLI	Drafting of Subsection 4.2.2 on organizational capabilities as the channel (13 August 2026), the largest cluster: 22 of its 26 studies are cited individually. All passages read directly from the source PDFs with pdfplumber; 42 quoted passages verified programmatically before entering footnotes (word sequence on the cited PDF page); printed page numbers from the review index’s per-document offsets, abstract references for the two studies with unusable pagination. Sixteen BibTeX entries generated from the frozen Scopus export. The blind citation check (Gemini 3.1 Pro) ran three times: the first pass flagged 17 of 34 sites, leading to five precision corrections in the prose and sample-statement extensions in seven footnotes; the final pass leaves five flags, all word-sequence echoes on construct names, sample descriptors, or the quantified threshold, each with Gemini judging the paraphrase fully supported — for the author’s batch decision. Text to be reviewed by the author.

Section		Tool		Description
Results thesis, 4.2.3)	(syn- Section	Claude (Anthropic) Antigravity (Google)	Code + CLI	Drafting of Subsection 4.2.3 on leadership, governance, and management (13 August 2026): 12 of the cluster’s 13 studies cited individually. Seventeen quoted passages verified programmatically on the cited PDF pages before entering footnotes; one wording assumption was falsified by the check and corrected to the page’s actual wording. Printed pages from page headers and the review index; abstract references for the three studies without usable pagination. Eight BibTeX entries generated from the frozen Scopus export. The blind citation check (Gemini 3.1 Pro) ran three times: seven initial flags led to four precision corrections in the prose and three footnote extensions with sample statements. Final state across the whole chapter: 9 of 46 citation sites flagged, all word-sequence echoes on construct names, sample descriptors, directive names, or the quantified threshold, each judged fully supported by Gemini; five already decided by the author, four awaiting the author’s batch decision.

Section		Tool		Description
Results thesis, 4.2.4)	(syn- Section	Claude (Anthropic) Antigravity (Google)	Code + CLI	Drafting of Subsection 4.2.4 on firm characteristics and market position (13 August 2026): all 16 studies of the cluster cited individually, including the two benchmark studies Babina et al. and Lui et al. Nineteen quoted passages verified programmatically on the cited PDF pages; seven BibTeX entries generated from the frozen Scopus export, three proper nouns brace-protected by hand. The blind citation check (Gemini 3.1 Pro) ran three times: thirteen initial flags led to ten precision corrections in the prose (claims cut back to what the abstracts state) and one footnote extension. Final state across the whole chapter: 15 of 62 citation sites flagged, every one a word-sequence echo on construct, sample, or metric names with Gemini judging the paraphrase fully supported; nine decided by the author (recorded durably), six awaiting the author’s batch decision.
Results thesis, 4.2.5)	(syn- Section	Claude (Anthropic) Antigravity (Google)	Code + CLI	Drafting of Subsection 4.2.5 on environmental conditions (13 August 2026): all 15 studies of the cluster cited individually, five re-cited with passages already verified in earlier subsections. Eighteen quoted passages verified programmatically on the cited PDF pages; six BibTeX entries generated from the frozen Scopus export. The blind citation check (Gemini 3.1 Pro) ran four times: ten initial flags led to seven precision corrections in the prose (two multi-study sentences split, two narrative frames removed, three wordings tightened) and six footnote extensions. Final state across the whole chapter: 22 of 77 citation sites flagged, every one a word-sequence echo on construct, sample, directive, or metric names with Gemini judging the paraphrase fully supported; fifteen decided by the author, seven awaiting the author’s batch decision.

Section		Tool		Description
Results thesis, 4.2.6)	(syn- Section	Claude (Anthropic) Antigravity (Google)	Code + CLI	Drafting of Subsection 4.2.6 on dose, credibility, and timing (13/14 August 2026), completing Section 4.2: all 14 studies of the cluster cited individually, six re-cited with previously verified passages. Twelve quoted passages verified programmatically; one extraction artifact resolved by reading the printed page image directly. Four BibTeX entries generated from the frozen Scopus export. The blind citation check (Gemini 3.1 Pro) ran three times: six initial flags led to five precision corrections and two footnote extensions. Final state for the finished chapter: 27 of 91 citation sites flagged, all word-sequence echoes on construct, sample, or term names with Gemini judging every paraphrase fully supported; fifteen decided by the author, twelve awaiting the author’s batch decision.
Results (Chapter 4)	(Chap- ter 4)	Claude (Anthropic) Antigravity (Google)	Code + CLI	Blind section review of the finished Results chapter (14 August 2026): Gemini 3.1 Pro received only the chapter prose and the fact sheet and flagged 4 findings — one genuine consistency error (two subsections restated the same dose findings nearly verbatim; fixed by cross-referencing instead of repeating) and three style patterns (a section-closing formula repeated four times, reduced to one instance; two stock idioms replaced). The author approved all fixes. Final state: 88 citation sites, 26 flagged, all word-sequence echoes with Gemini judging every paraphrase fully supported, all covered by the author’s recorded decisions. Chapter 4 thereby passed all three cross-check gates.

Section		Tool		Description
Theory tions 2.1–2.5)	(Sec-	Claude (Anthropic) Antigravity (Google)	Code + CLI	Citation re-check of the theory chapter on the migrated backend (14 August 2026): alongside the 7 known word-sequence flags, the migrated script surfaced the paraphrase-fidelity verdicts, including one genuine misreading (Barney 1991, p. 106 — the prose had inverted the passage’s point that the VRIN characteristics alone do not make an attribute a resource). Nine precision corrections applied to the prose and four footnotes extended with sample or mechanism passages read from the PDFs (printed pages verified against running heads). Final state: 8 of 25 citation sites flagged — 6 word-sequence echoes on construct or term names with the paraphrase judged fully supported, one definition-not-finding note, one simplification of Barney’s fourth condition — all decided by the author in one batch session (14 August 2026, recorded durably). The theory chapter thereby passed all three cross-check gates.
Theory tions 2.1–2.5)	(Sec-	Claude (Anthropic) Antigravity (Google)	Code + CLI	Blind section review of the theory chapter (14 August 2026, Gate 3 of the cross-check workflow): Gemini 3.1 Pro received only the chapter and the fact sheet and flagged 2 findings, no number contradictions. One consistency flag accepted and fixed (a transition promised an illustration of the productivity dip but led into a study finding a clear gain; reworded to announce the contrast the paragraph actually draws), one style flag rejected as void (the flagged sentence is a verbatim source quote inside a footnote). Both resolutions confirmed by the author in the same batch session.

Section	Tool	Description
Appendix (coding table)	Claude Code (Anthropic)	Full-text verification of all appendix abbreviation resolutions (14 August 2026): every acronym resolution in the 23 reworded cells checked against its source PDF. Nineteen confirmed exactly; four precision corrections applied (Leoni’s construct completed to “manufacturing firm performance”, Sullivan’s mediator to “adaptive response to market changes”, Bin-Nashwan’s moderator path to “green intellectual capital”, and the Sun 2022 cell extended to name the two null value paths the rewording had silently dropped). Table regenerated with its self-checks passing.
Discussion	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of the full Discussion chapter (14 August 2026): research question answered separately for the two constructs, evidence interpreted through the four theory lenses, implications, limitations, future research. 25 citation sites, of which 22 re-cite previously verified passages and 3 cite Vaillant et al. (2025), the first empirical application of situated AI theory, read directly from the PDF (early view, section-based citations). The blind citation check ran three times: 18 initial flags led to six sentence splits, four footnote extensions, and six precision corrections (causal wording weakened to the sources’ correlational findings, theory claims separated from corpus findings, own-synthesis framing separated from cited claims). Final state: 10 of 25 sites flagged, all word-sequence echoes on construct, sample, or threshold names judged fully supported, all decided by the author in one batch session (14 August 2026, durable log). All corpus counts from the fact sheet.

Section	Tool		Description
Discussion	Claude (Anthropic) Antigravity (Google)	Code + CLI	Blind section review of the Discussion chapter (14 August 2026, Gate 3; footnotes stripped before submission): Gemini 3.1 Pro flagged 7 findings — no number contradictions, no consistency errors, seven style patterns. Six fixes applied; one flag rejected because the parallelism is the source’s own summary rule. All resolutions confirmed by the author (14 August 2026). The Discussion chapter thereby passed all three cross-check gates.
Introduction	Claude (Anthropic) Antigravity (Google)	Code + CLI	Drafting of the Introduction (14 August 2026) along the four mandated moves: Kemp’s theoretical puzzle as the anchor (two new passages read column-wise from the PDF), prior research summarized through four already-verified corpus passages, the research question stated explicitly, the outlook covering protocol and chapter structure. Citation check ran twice (two genuine fixes: a footnote extension for the sample descriptor, association wording restored at Babina; remaining flags are fully supported term echoes). Blind section review flagged 3 findings, all fixed: search-window wording disambiguated, two antithesis constructions rewritten plainly. All resolutions and flags decided by the author (14 August 2026). The Introduction thereby passed all three cross-check gates.

Section		Tool		Description
Conclusion Abstract	+	Claude (Anthropic) Antigravity (Google)	Code + CLI	Drafting of the Conclusion chapter and the Abstract (14 August 2026), both citation-free summaries with all counts from the fact sheet. The blind section review ran over both jointly and flagged 4 findings, all fixed — including one genuine number-attribution error (corpus-wide counts phrased as construct-specific ones), a combinations claim cut back to the configurational evidence, and two antithesis constructions rewritten plainly. All four resolutions reviewed individually and accepted by the author (14 August 2026).
Method (interim draft)	(in-	Claude Code (Anthropic)		First supervisor draft (29 July 2026): Claude assembled the standalone compile of the interim draft — TOC skeleton, the full Method chapter, and the coding-table excerpt (the 25 studies in the highest-ranked journals plus three AJG 2 examples). Citation-verification footnotes are suppressed in the compiled draft (working files keep them); the drafted theory chapter and the List of Aids were left out of that round — the former to keep the review load at one chapter, the latter pending the supervisor’s guidance on how AI use should be documented.

Section	Tool	Description
Method	Claude Code (Anthropic)	Drafting of the full Method chapter (29 July 2026): protocol prose assembled from the frozen method documentation (query wording, run date, $AJG \geq 2$ rule, screening chain $432 \rightarrow 174 \rightarrow 81 \rightarrow 67$, E-code catalogue, dual-coding procedure with agreement statistics and the harmonization pass, pagination rule for the seven studies without printed page numbers). PRISMA-style flow diagram generated by <code>corpus/make_prisma.py</code> from the screening-record counts, with checksum self-tests. One anchor citation (Rojon et al. 2021, p. 196) read column-wise from the PDF and footnoted per convention; a candidate passage on p. 195 was identified as a Tranfield quotation inside Rojon and rejected as a secondary source. The three benchmark studies are cited as entities without page numbers (no content claim). Human-voice pass applied. Text reviewed and edited by the author before sending.

Section	Tool	Description
Method (author's rewrite)	Claude Opus 5 (Anthropic) + AntigraVity CLI (Google)	Reintegration of the author's own rewrite of the Method chapter (17 August 2026): the author rewrote the whole chapter in their own words outside LaTeX; Claude Opus 5 checked the rewrite against the frozen method documentation and the fact sheet — every count, date, and PRISMA number confirmed unchanged — and restored what the Word version could not carry: section structure, the query block, the PRISMA figure, the four cross-references (all verified against the compiled numbering), and the three verification footnotes with their page numbers (Rojon p. 196, Chatterjee p. 209, Babina p. 1). Both example citations were moved back behind their claims, since sentence-initial placement had been flagged by the citation check on 13 August. Garbled sentences and typos corrected; the ISSN basis of the AJG step, the patent-based AI measure, and the eligibility-check caveat restored. Blind citation check re-run on the finished chapter: 3 of 3 sites pass, 0 flagged.

Section	Tool	Description
Results (coding audit)	Claude Fable 5 (Anthropic)	Evidence dossier for the author’s final audit of the coding table (18–19 August 2026): for each of the 67 coded rows, Claude read the frozen full text and extracted the verbatim source passages backing every coded column — 1,019 quotes in total, each tagged with the columns it evidences and its printed page (section references for the seven studies without usable pagination). Every quote was then machine-verified by the script <code>corpus/coding_evidence.py</code>, independent of the extracting model, in the same two stages and with the same calibrated threshold as the library citation check (verbatim match in the extracted full text; word sequence on the real PDF page): 894 confirmed on the page, 125 marked for manual page inspection (in line with the known extraction false-alarm rate of roughly one in ten), none unresolved after 12 quotes were corrected to the source’s exact wording where the check caught transcription drift. The dossier’s summary flags 16 studies with notes for the author’s decision — including a country coding that contradicts the sample’s origin, one never-adjudicated direction code, and three papers that are internally inconsistent; no coded value was changed by the tool. For manual spot-checking, every quote additionally carries a Ctrl+F search string copied from the PDF’s own text layer, so it can be found in the viewer even where the readable quote differs from the PDF internals; 1,017 of 1,019 quotes carry one, the two exceptions are marked for visual checking. The appendix tables were regenerated from the coding table and diffed byte-identical against the committed version. All flags remain the author’s decisions.

Section	Tool	Description
Appendix (coding table)	Claude Fable 5 (Anthropic)	Condensed reader-facing appendix table (19 August 2026, author’s request): a second rendering layer (corpus/appendix_condensed.tsv, 261 cells) replaces coder-fidelity cells with plain-language short versions — samples reduced to n + unit (+ period); estimator names, response rates, recruitment channels, and vendor lists dropped; abbreviations spelled out; AI measures given a plain gloss; conditions keeping every condition and its role but losing path anatomy and coefficients; key findings reworded only where they carried jargon (14 of 67). A Notes column with short evidence tags derived from quality_notes was trialled in Table A.2 and removed again the same day at the author’s decision (the tags remain in the layer file, unprinted). Style anchored on five cells the author approved before the rollout; the Outcome/Perf/CA pairing kept unchanged per the author’s instruction. The coded values in corpus/coding_table.csv are untouched; two machine self-checks guard the layer (no number may appear in print that is absent from the coded row; no coder shorthand may survive). Every changed cell documented before/after in corpus/appendix_condense_review.md for the author’s review.

Section	Tool		Description
Results (coding audit)	Claude (Anthropic)	Fable 5	Cell-audit viewer built into the local library reader (19 August 2026): a keyboard-driven view that walks all 826 non-empty cells of the coding table study by study in the printed tables’ column order; for each cell it shows the coded value and its machine-verified evidence quotes, opens the source PDF at the quoted page, and highlights the passage. One key records “cell OK”, one records a flag with a note; judgments land append-only in corpus/author_audit.csv — the coded values are never touched by the tool. The resulting file documents the author’s cell-by-cell verification of the coding table against the sources.
Results (coding audit)	Claude (Anthropic)	Fable 5	First correction pass from the author’s cell audit (19 August 2026): 27 author flags from corpus/author_audit.csv verified against the evidence quotes and 18 author-approved cell corrections applied to the coding table — among them two sample-country corrections (S08 to USA per the paper’s own limitation statement; S21 to Europe/France+UK, the coded country had been the authors’ affiliation), one effect direction (conditional to positive for a study with a robust average main effect that had never been individually adjudicated), measurement precision for the country-level venture-capital proxy plus a previously missing firm-size moderator, and sample-unit corrections (responses vs. firms). Check-requests that verified as already correct were answered and left unchanged. All downstream artifacts regenerated with their self-checks (appendix tables, fact sheet, figures, dossier); the corpus statistics changed to 39 positive / 19 conditional and the five prose sentences citing the old counts were updated. Every change traces to a flag row in the audit file.

Section	Tool	Description
Method (author's rewrite, 2nd pass)	Claude Fable 5 (Anthropic) + Antigravity CLI (Google)	Reintegration of the author's second manual rewrite of the Method chapter (19 August 2026): the author reworded the chapter once more outside LaTeX; Claude transcribed the prose 1:1 (only obvious typos and extraction artifacts corrected, each listed for the author), restored the LaTeX apparatus (section structure, query block, PRISMA figure, cross-references, the three verification footnotes), and verified every count, date, and PRISMA number against the frozen record — all unchanged. Discrepancies were reported to the author instead of being fixed silently. Blind citation check re-run (Gemini 3.1 Pro): 3 of 3 sites flagged — one substantive wording flag and two artifacts of citation placement — all awaiting the author's decisions.

Section	Tool	Description
Results (coding audit)	Claude Fable 5 (Anthropic)	Second correction pass from the author’s cell audit (20 August 2026): all 62 open flags in corpus/author_audit.csv resolved with full-text verification against the source PDFs; briefs with page-cited evidence in corpus/audit_flag_resolutions.md. Nine flags had already been applied in the first pass, 14 verified as correct and left unchanged (each answered with the exact source passage), eight printed-table wordings shortened per the author’s dictation, and 24 coded cells corrected as proposals for the author’s batch review — among them one measurement definition read off the paper’s production function (sales-, not labor-based productivity), one effect direction changed to mixed after the profitability indicator proved null in every specification of the study, one mediation rationale corrected after a footnote revealed a supplementary direct-effects check, and two cells whose abstract-level claims were contradicted by the papers’ own coefficient tables. Three evidence quotes were re-pinned into the dossier where the audit had found claims without a backing passage. All downstream artifacts regenerated with their self-checks (fact sheet, appendix tables, figures, compiled thesis); the prose sentences citing the changed counts updated. All proposed values await the author’s batch approval.